



Masonic Cancer Center

UNIVERSITY OF MINNESOTA

The basis of genotoxicity, mutagenicity and carcinogenesis

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Masonic Cancer Center

University of Minnesota



A Cancer Center Designated by the
National Cancer Institute

Origins of Carcinogenesis

- genetic
- genomic instability
- epigenetic alterations
- viral
- hormones
- chemical

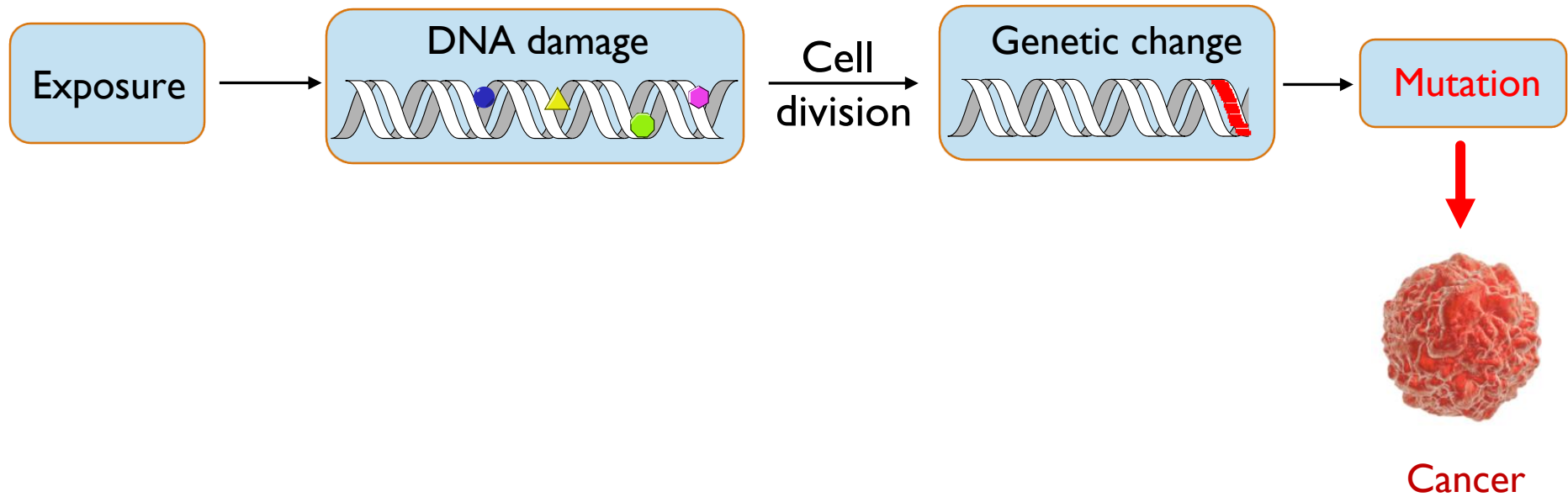
Origins of Carcinogenesis

- Cancers result from the **corruption** or **misuse** of genetic data (DNA) in a cell.

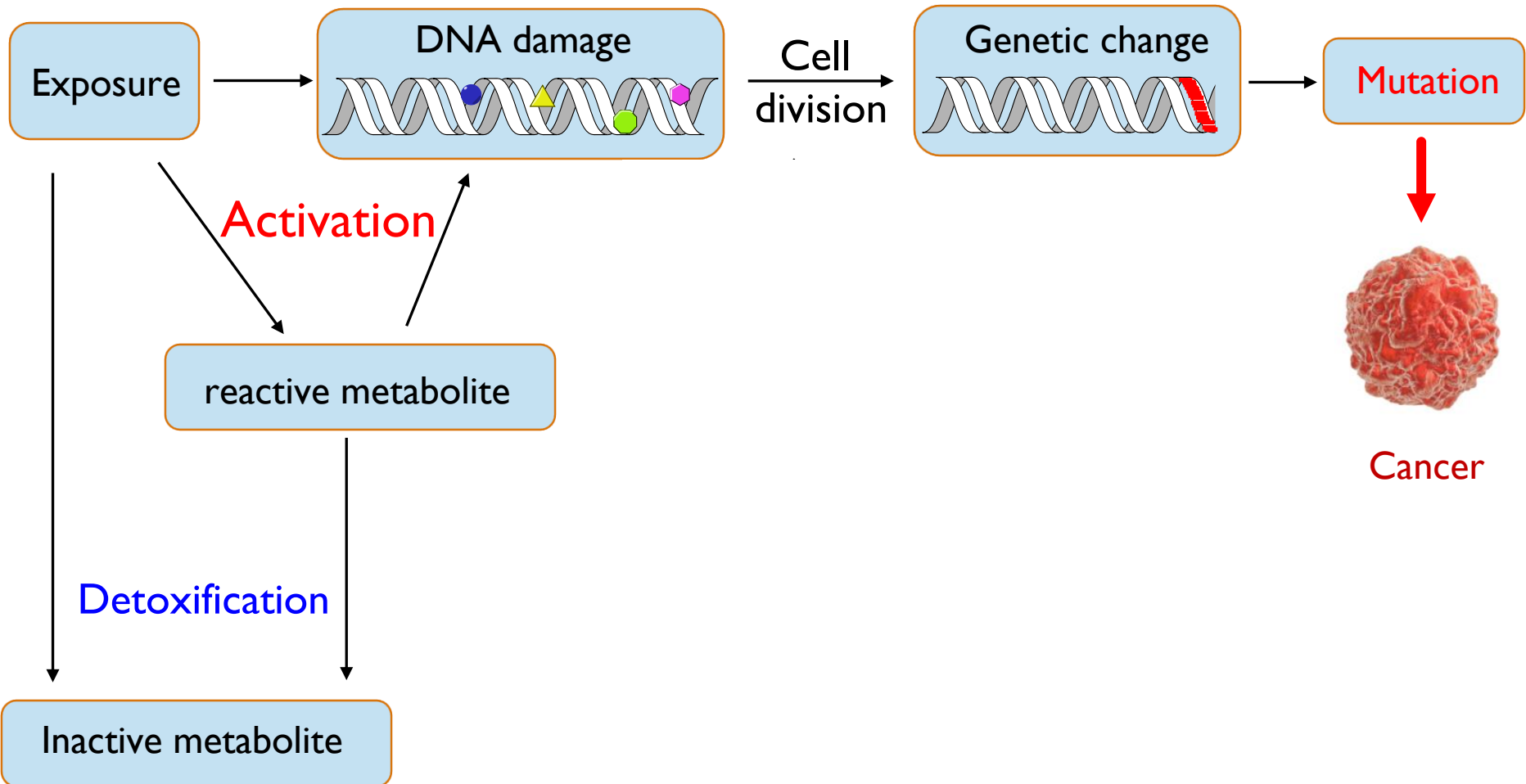
Origins of Carcinogenesis

- Cancers result from the **corruption** or **misuse** of genetic data (DNA) in a cell.
 - **Corruption** results from **changes** in DNA that can be small (mutations) or large (chromosomal abnormalities).
 - **Misuse** results from changes in **when** genetic information is accessed, and **which** portions are accessed (epigenetics).

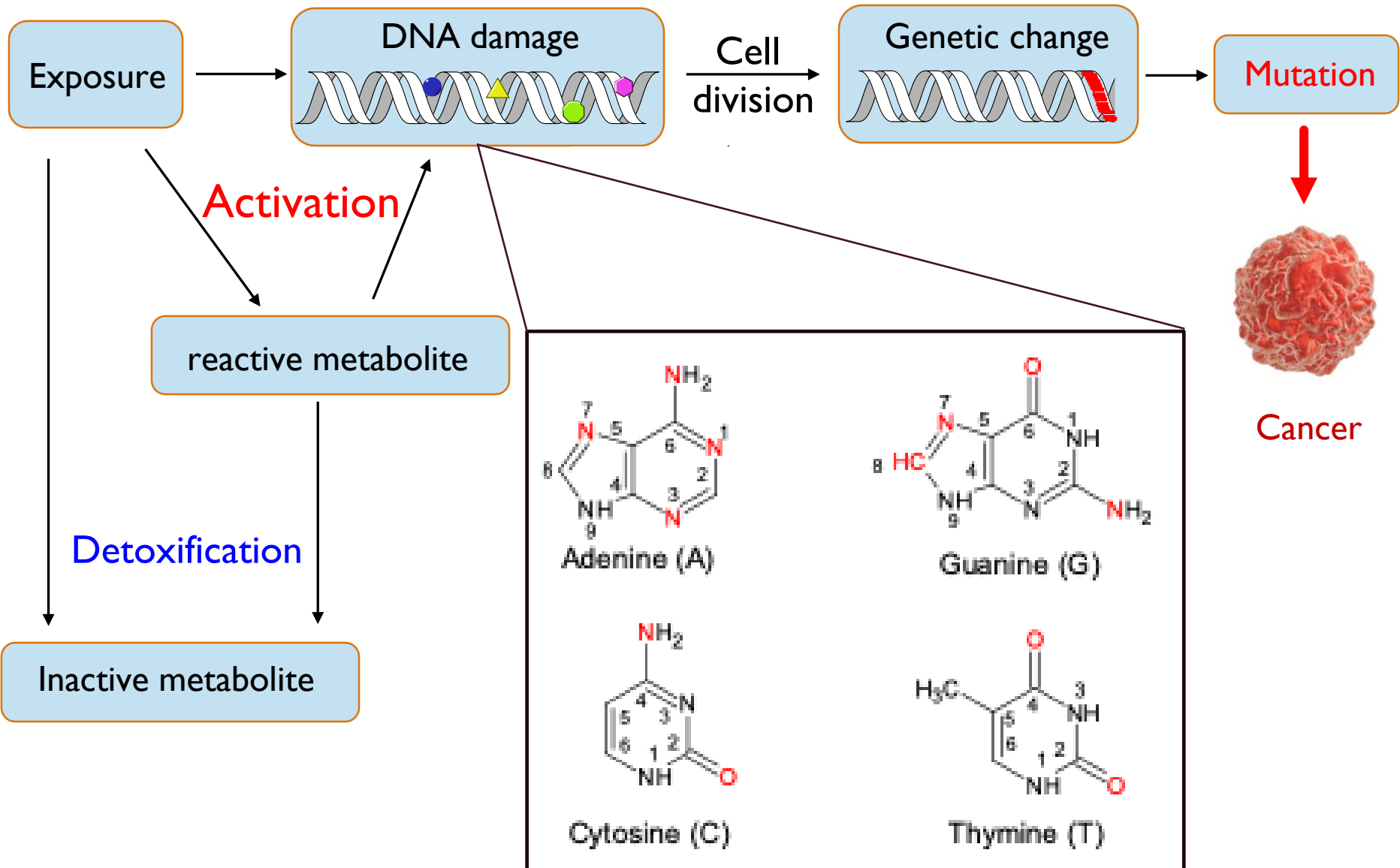
Chemical Carcinogenesis



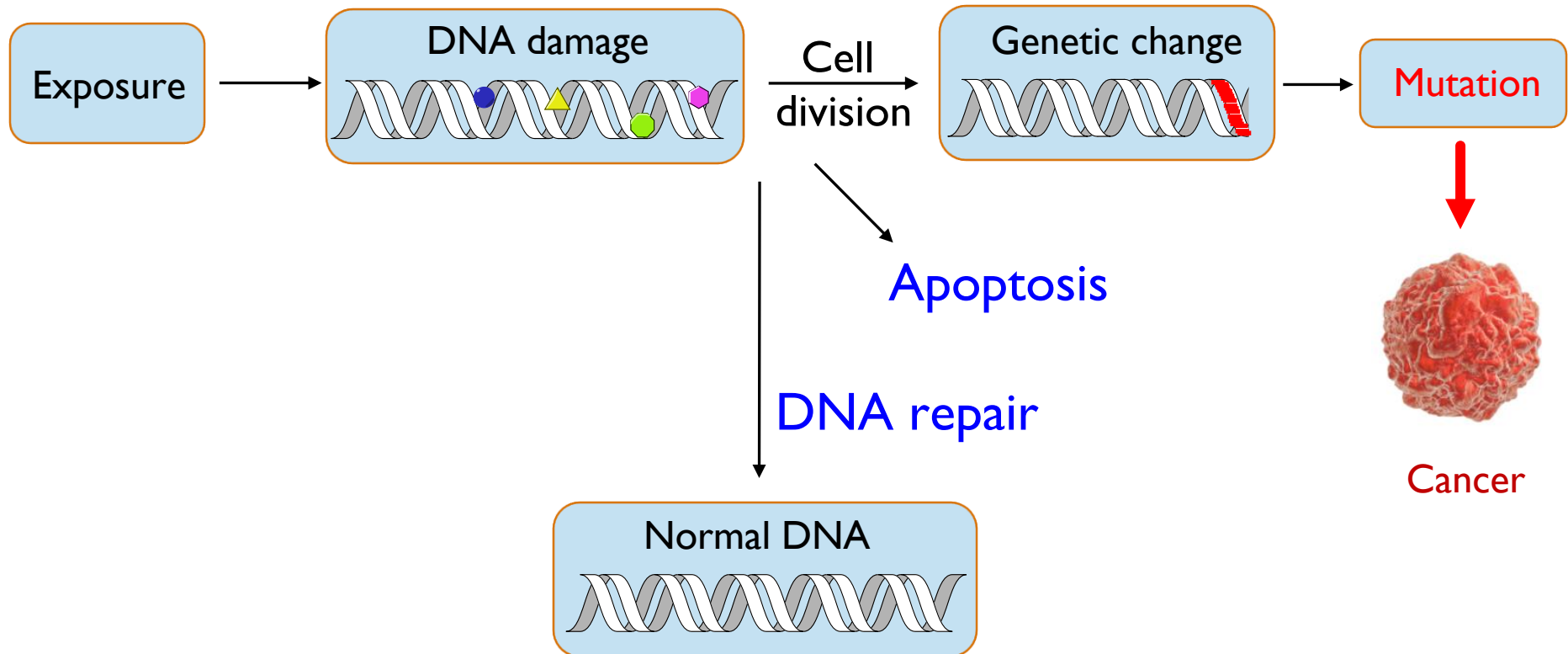
Chemical Carcinogenesis



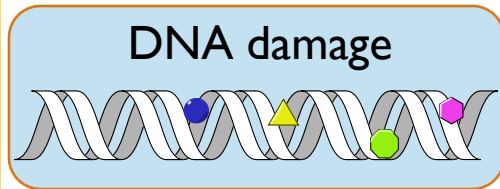
Chemical Carcinogenesis



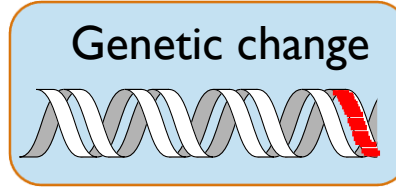
Chemical Carcinogenesis



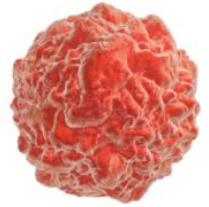
Chemical Carcinogenesis



Cell
division



Mutation



Cancer

Genotoxicity

Mutagenicity

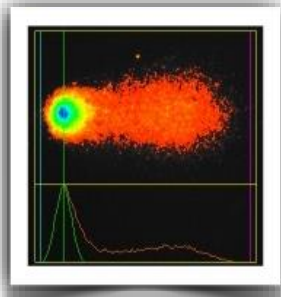
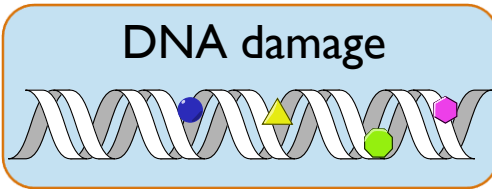
Carcinogenicity

Genotoxicity or Mutagenicity studies

Do not allow for the full evaluation of the mechanisms at a molecular level.

Therefore, we may need multiple studies and testing to assess carcinogenicity.

Traditional assessment of Chemical Carcinogenesis



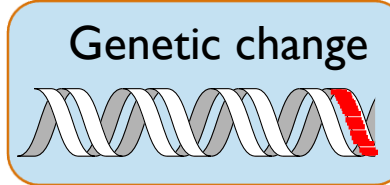
Comet Assay

ELISA 8oxo-dG

LC-MS DNA adducts

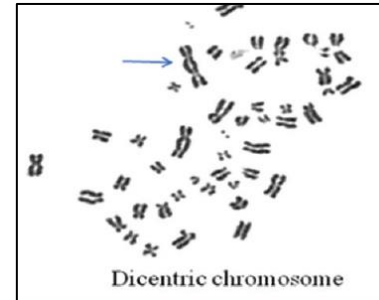
γ -2HAX

Cell
division

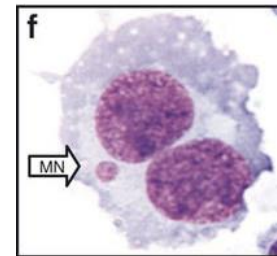


Mutation

Chromosomal
aberrations

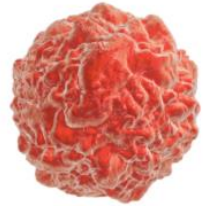
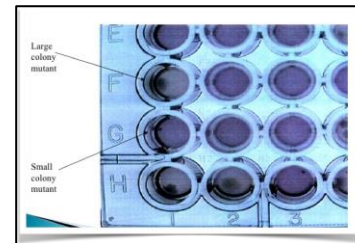


Micronuclei



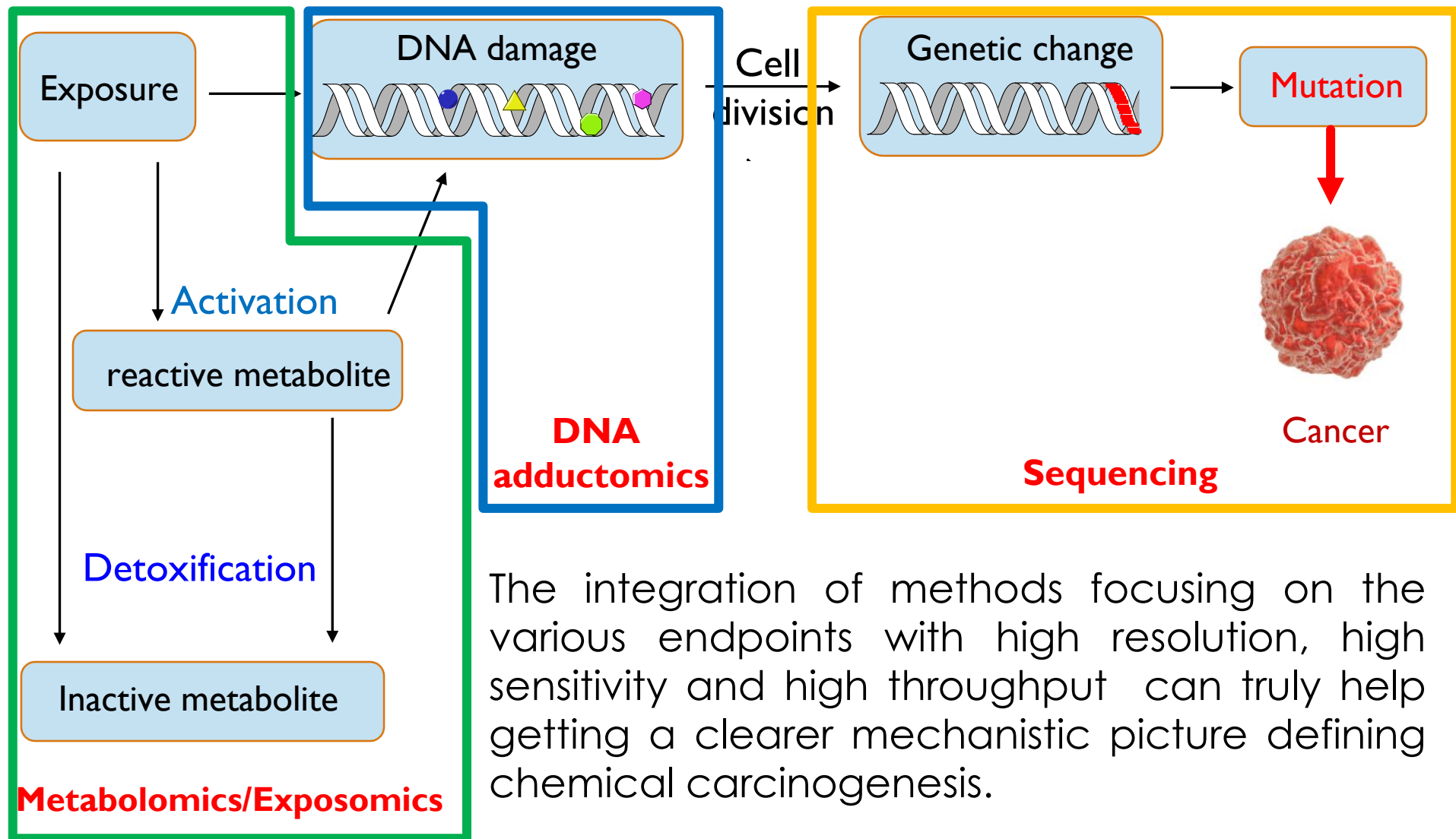
Mouse
lymphoma test

Ames test

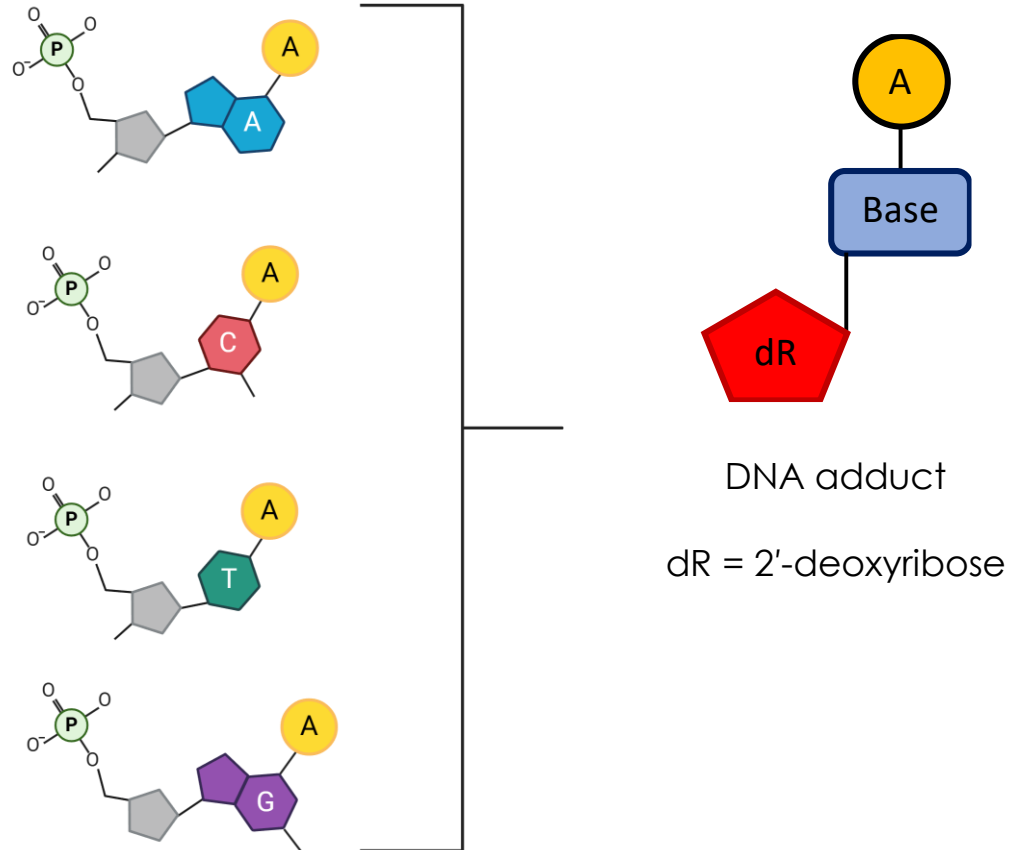


Cancer

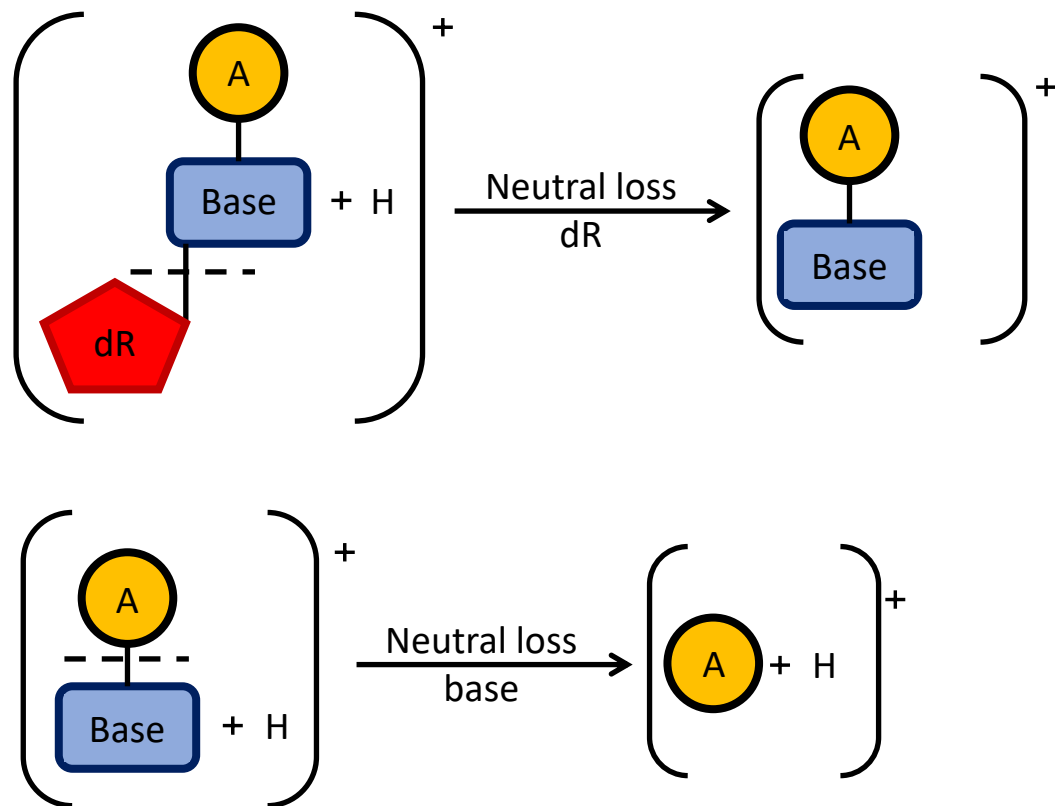
Novel approaches to study Chemical Carcinogenesis



DNA adductomics: “a top down approach”

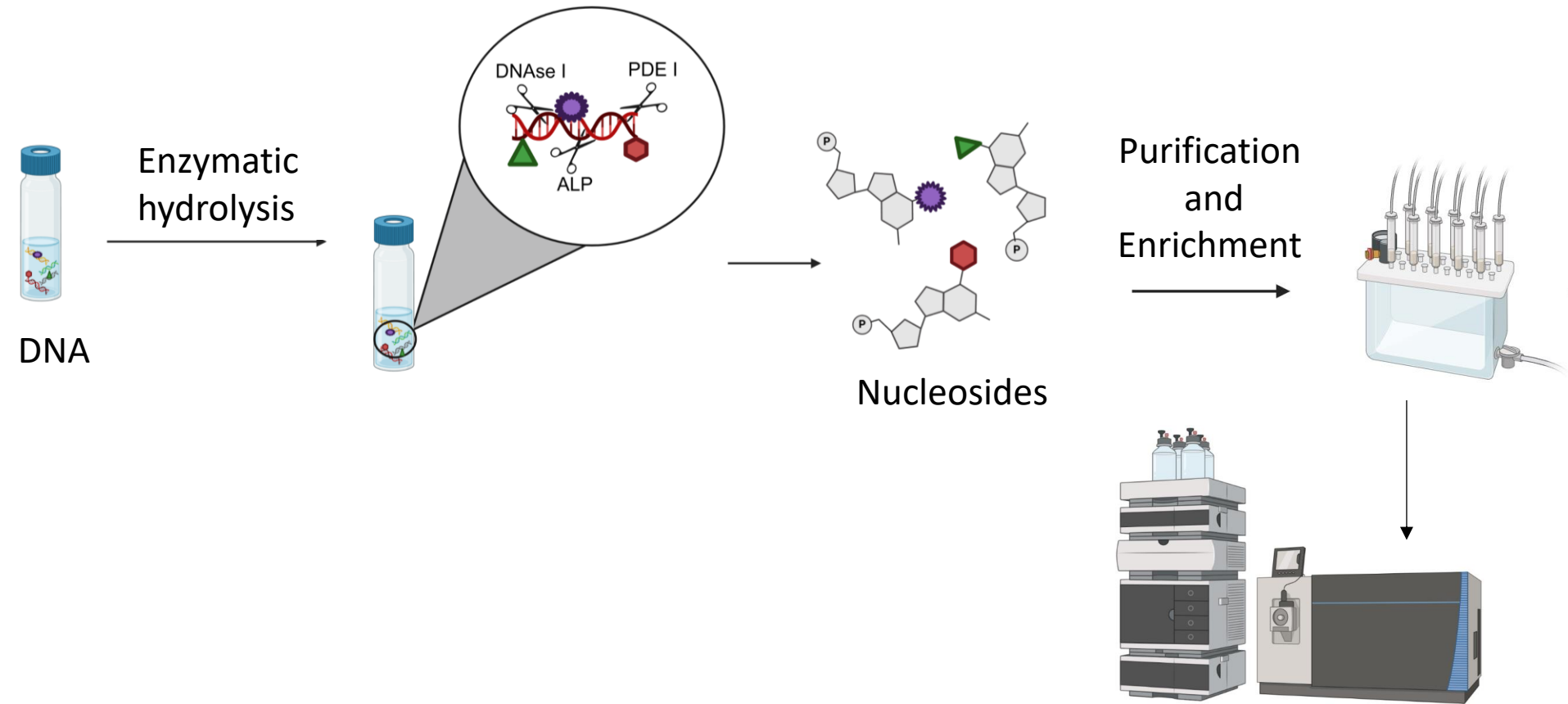


Taking advantage of diagnostic fragmentations

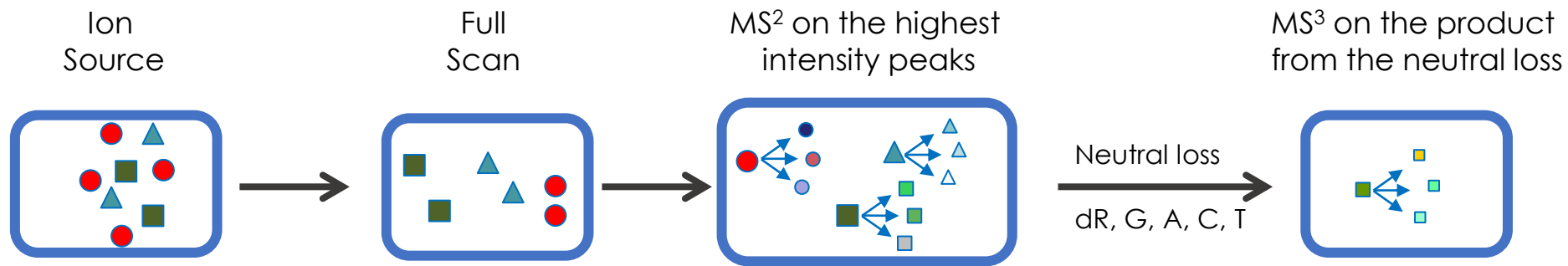


DNA adductomic approaches allow to chemically characterize the DNA damage resulting from a specific exposure.

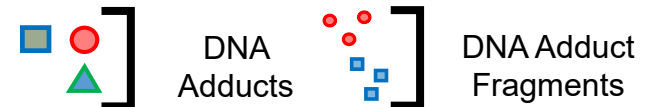
DNA Adducts Identification



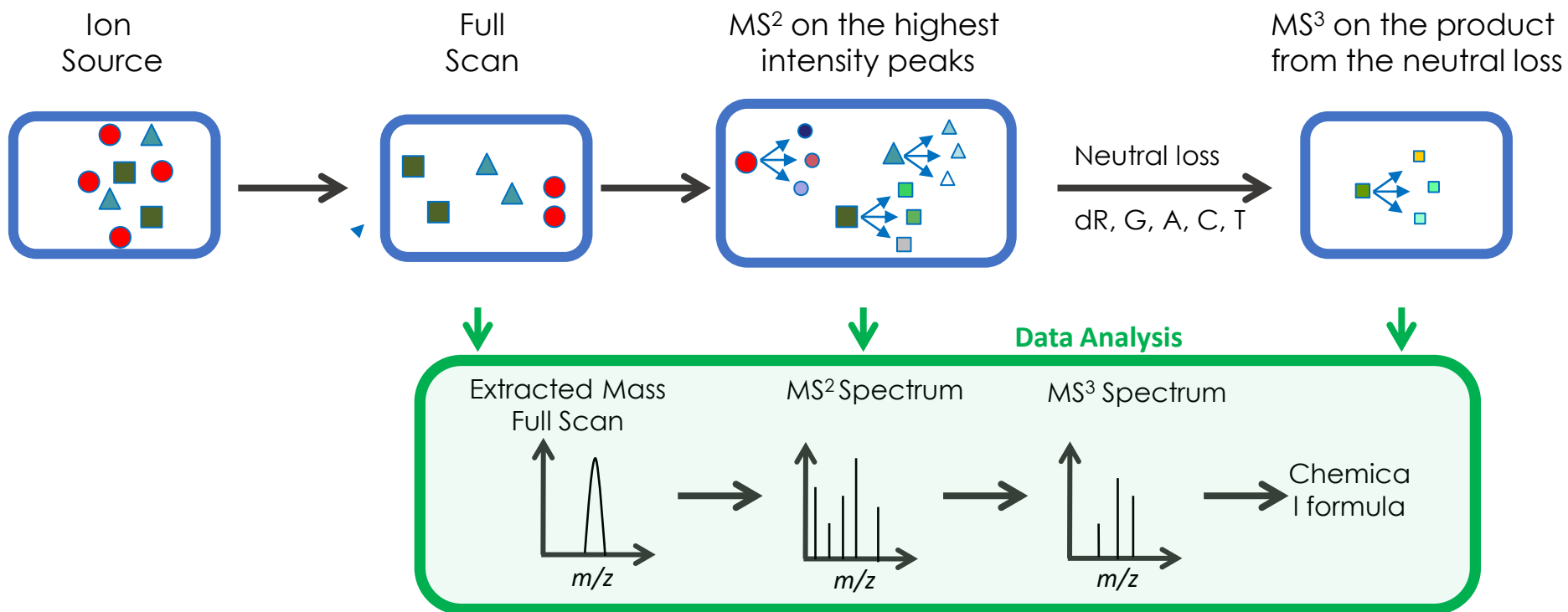
Data-Dependent Acquisition (DDA) Constant Neutral Loss/ MS^3

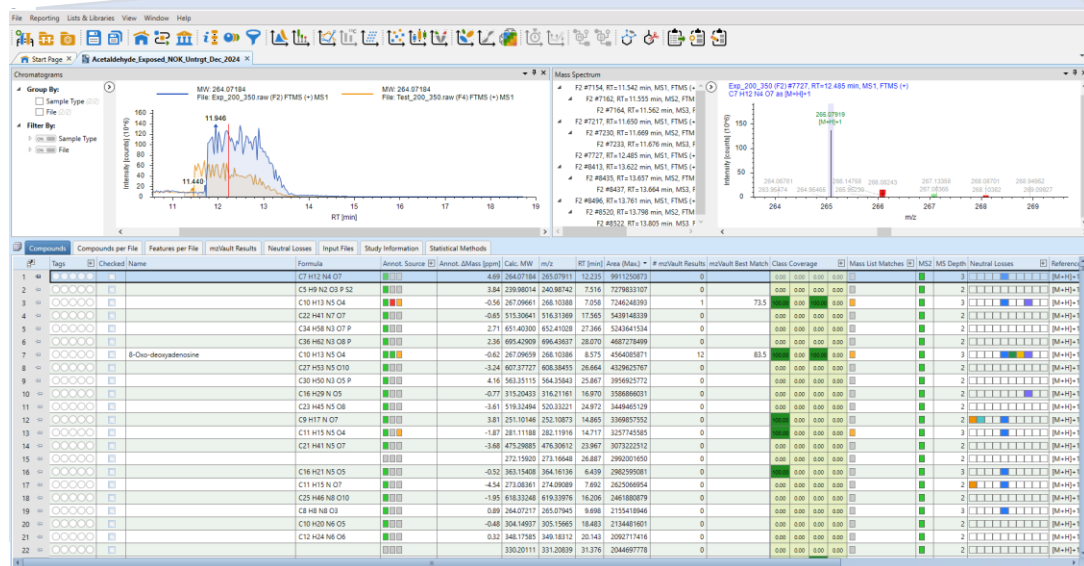


Multiple levels of fragmentation allow to gain structural information on the DNA adduct

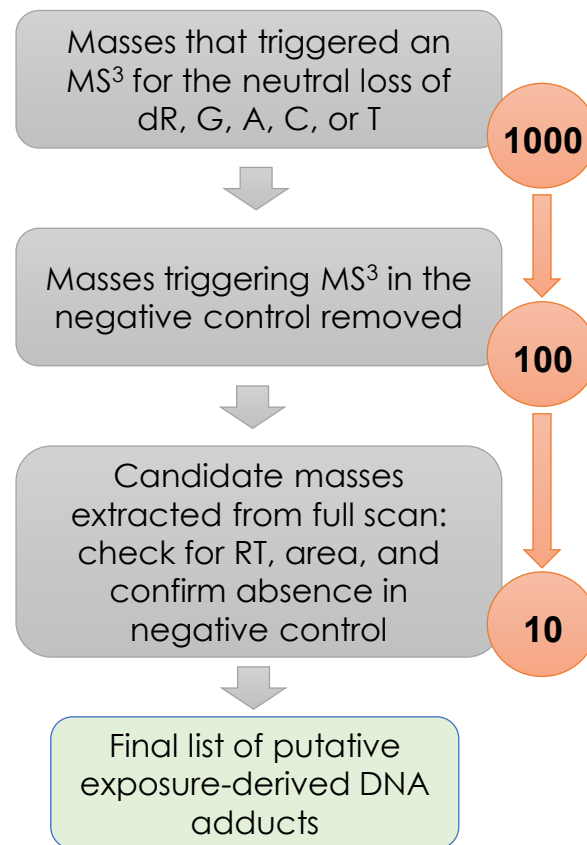


Data-Dependent Acquisition (DDA) Constant Neutral Loss/MS³



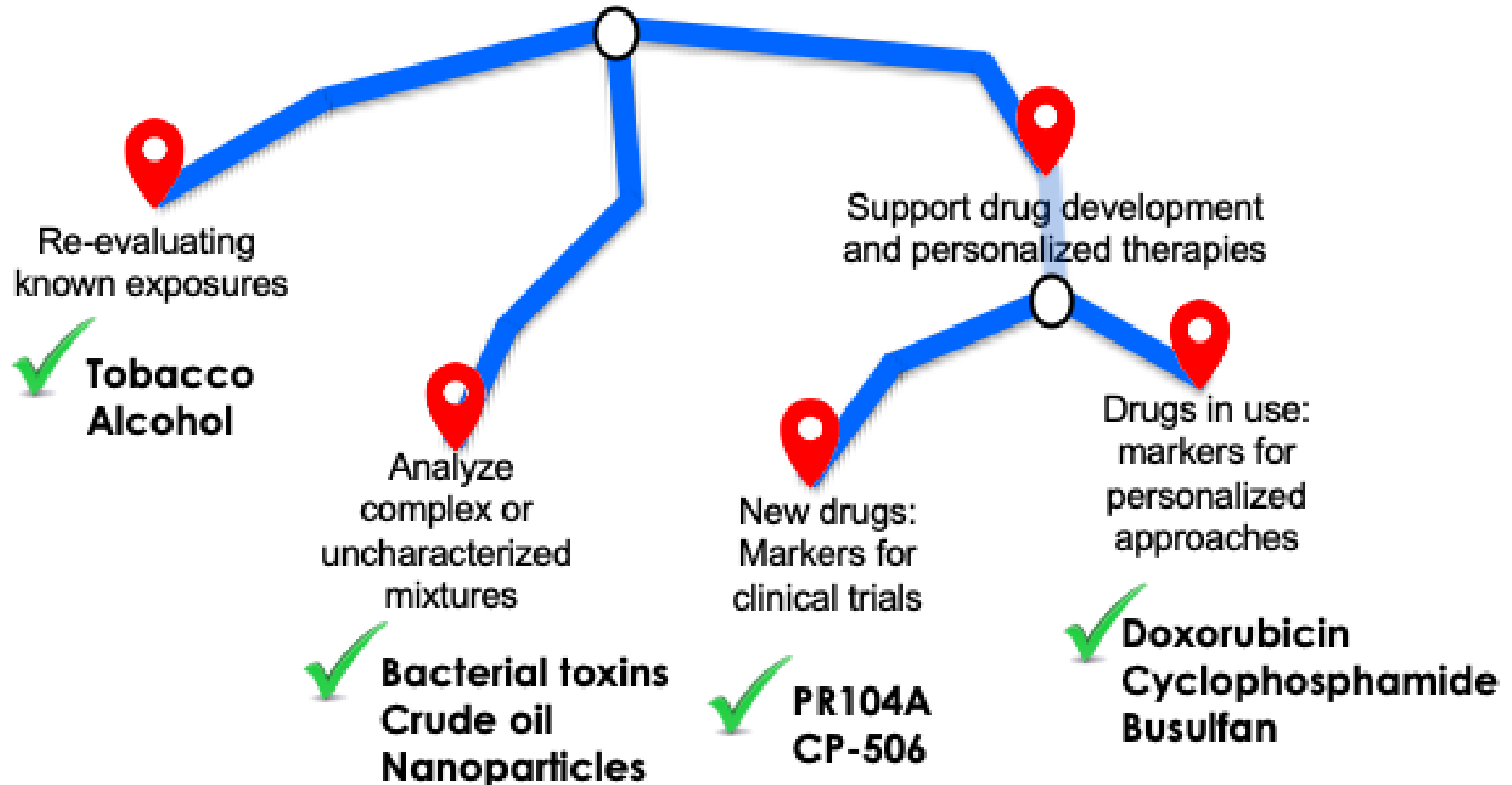


Careful Data Analysis

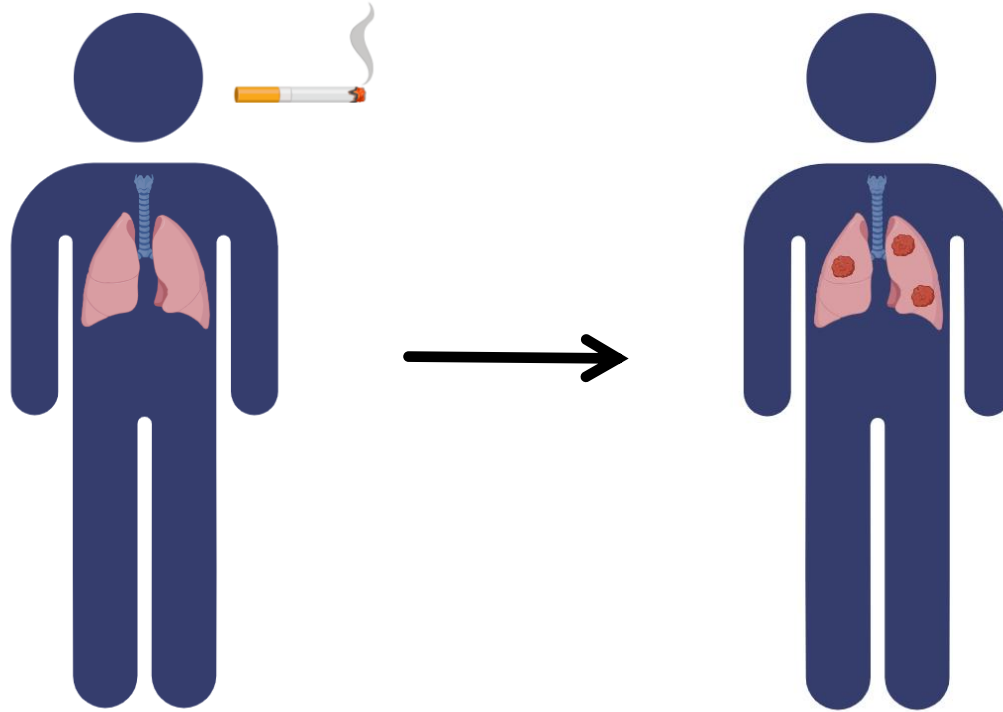


DNA adductomics analysis produces thousands of potential hits that are then filtered, vetted and scrutinized to generate a final list of DNA adducts.

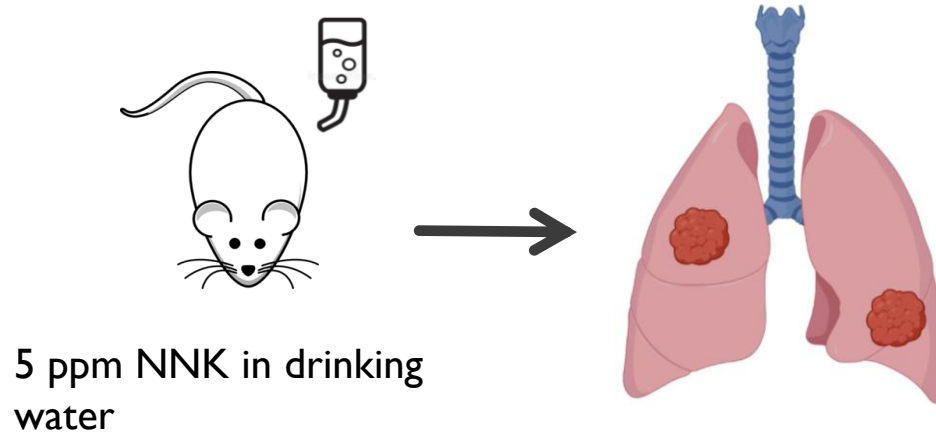
DNA adductomics applications



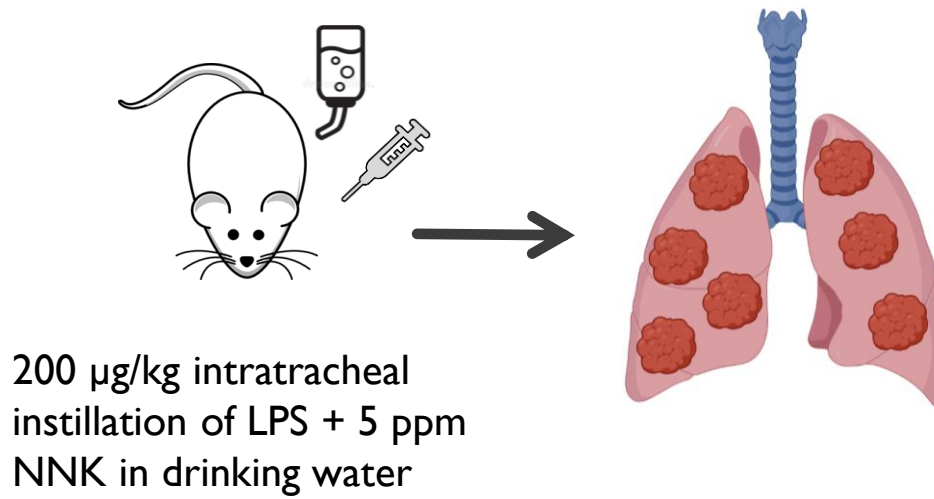
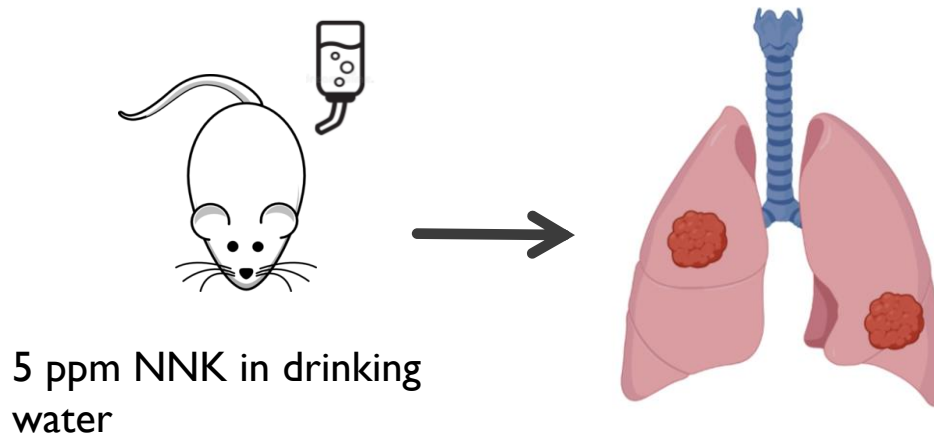
DNA adductomics applied to lung carcinogenesis



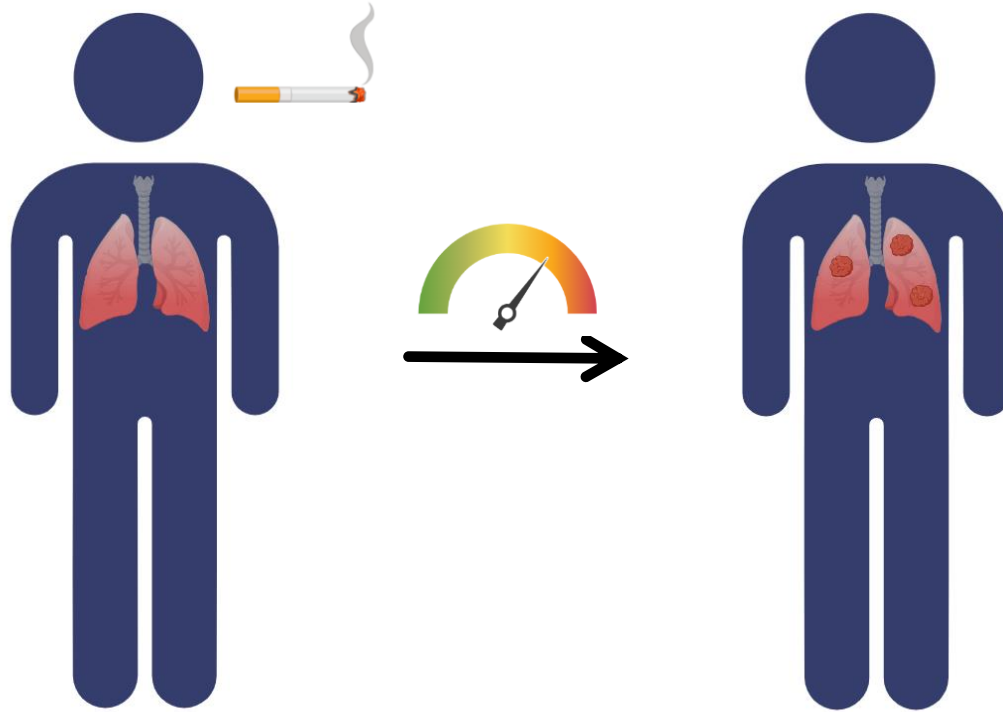
Lung carcinogenesis model using NNK



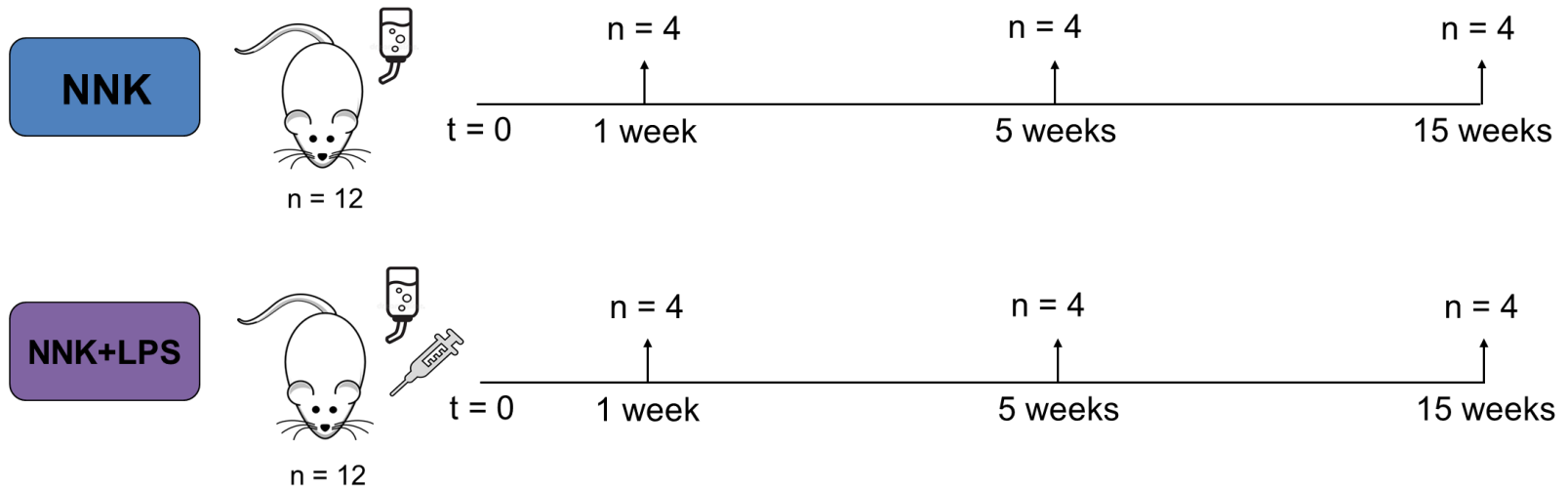
Lung carcinogenesis model using NNK



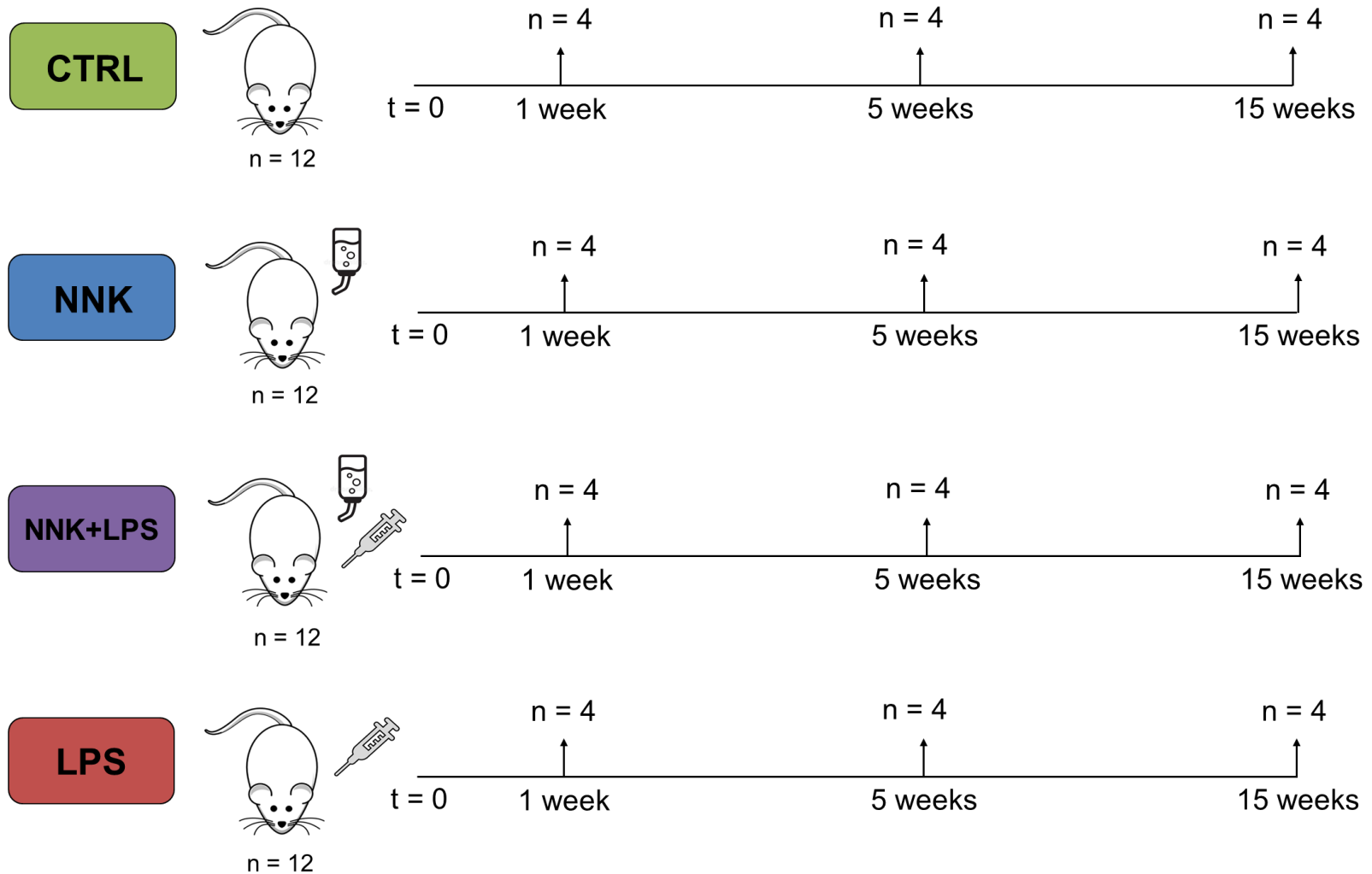
Inflammation increases the Risk of Lung Cancer



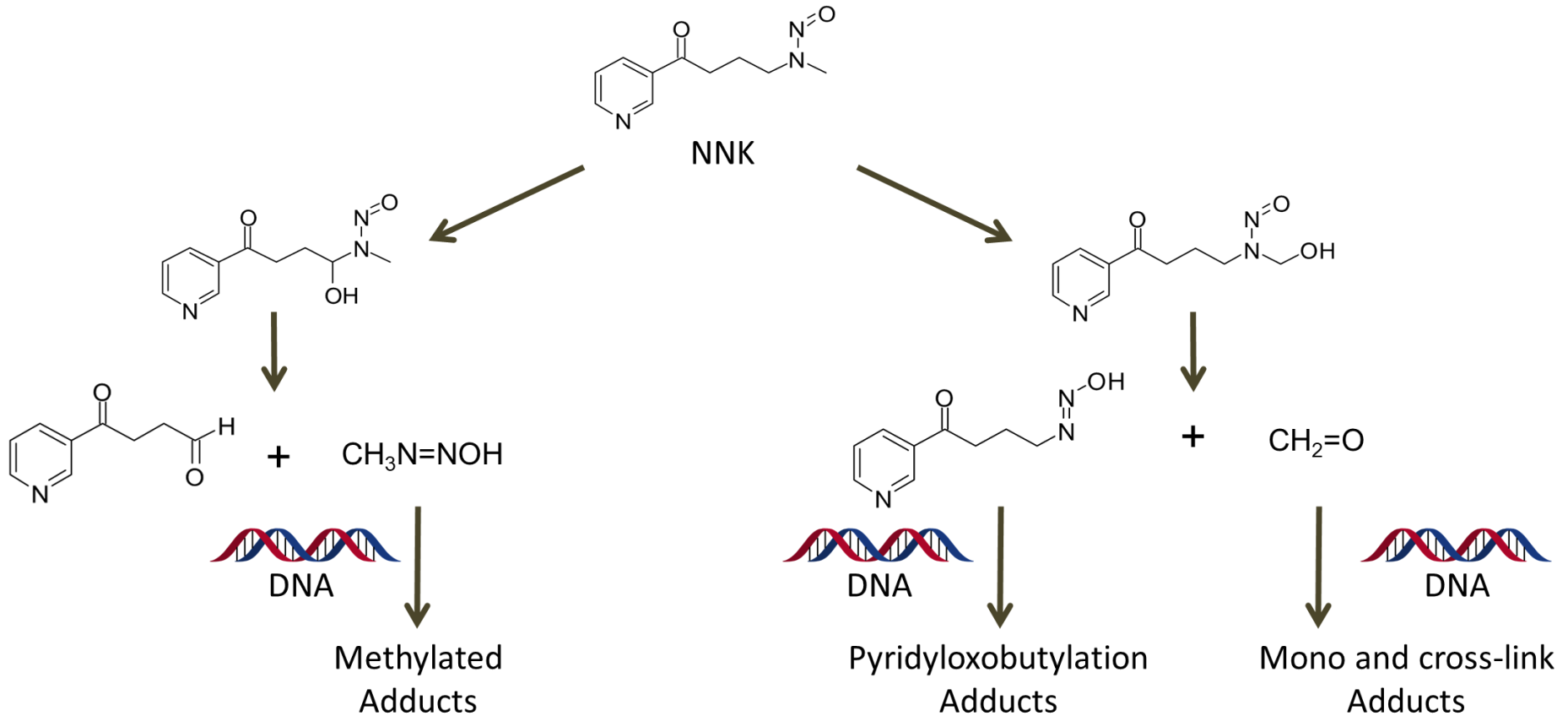
Our time course experiment



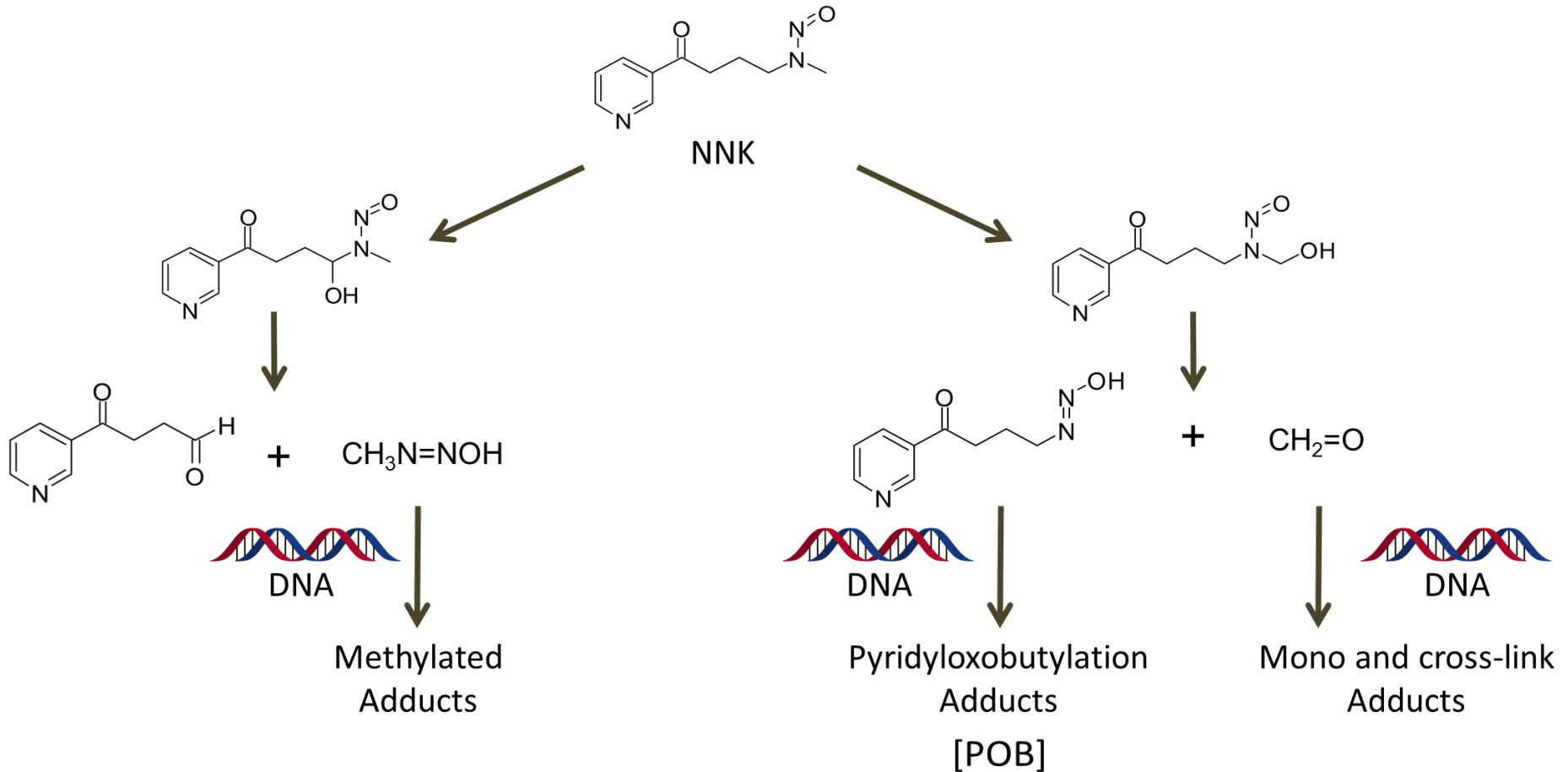
Our time course experiment



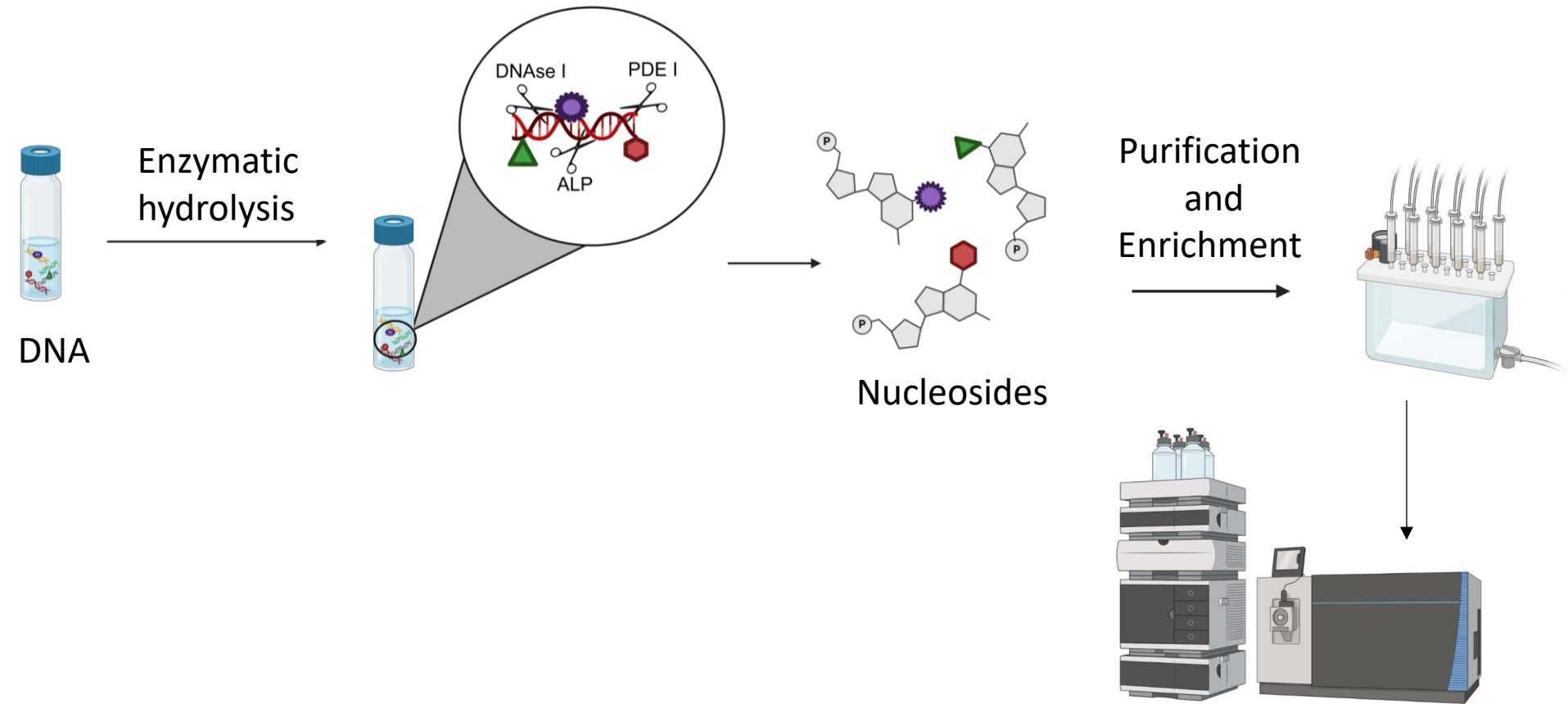
NNK metabolism



NNK metabolism

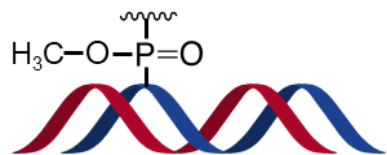
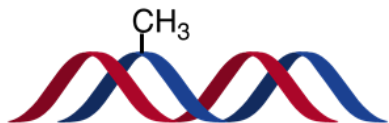


DNA Adducts Identification

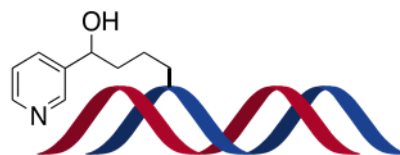
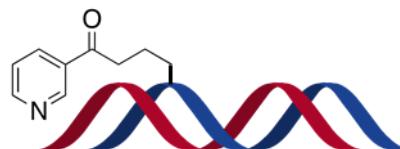


NNK-derived DNA adducts

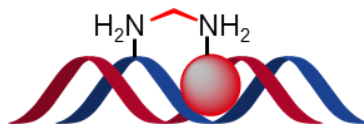
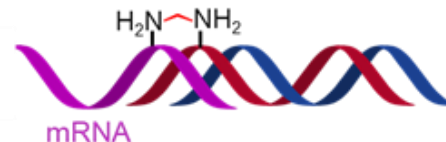
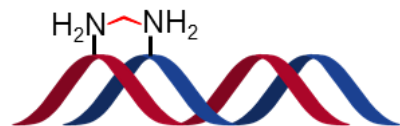
Methyl DNA Adducts



POB/PHB DNA Adducts

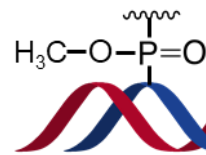
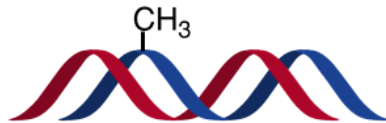


Cross-links (DNA-DNA, DNA-RNA, DNA-protein)



NNK-derived DNA adducts

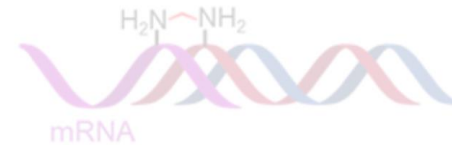
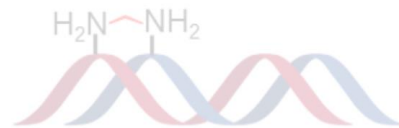
Methyl DNA Adducts



POB/PHB DNA Adducts

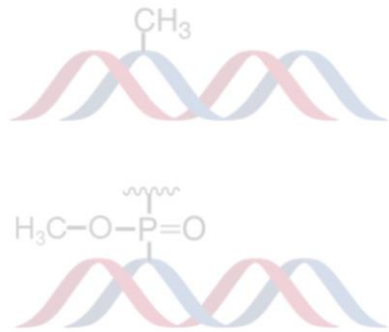


Cross-links (DNA-DNA, DNA-RNA, DNA-protein)

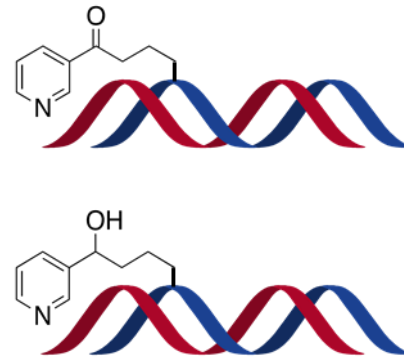


NNK-derived DNA adducts

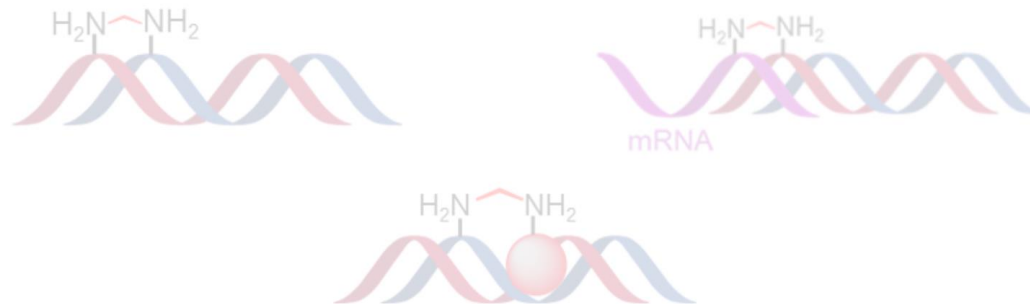
Methyl DNA Adducts



POB/PHB DNA Adducts

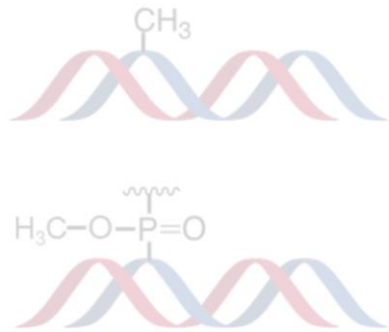


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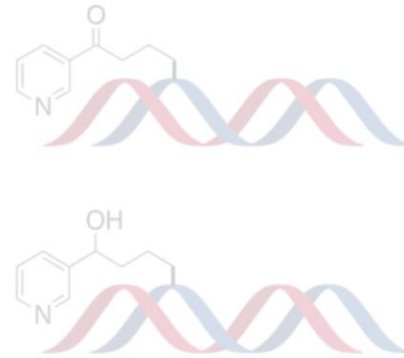


NNK-derived DNA adducts

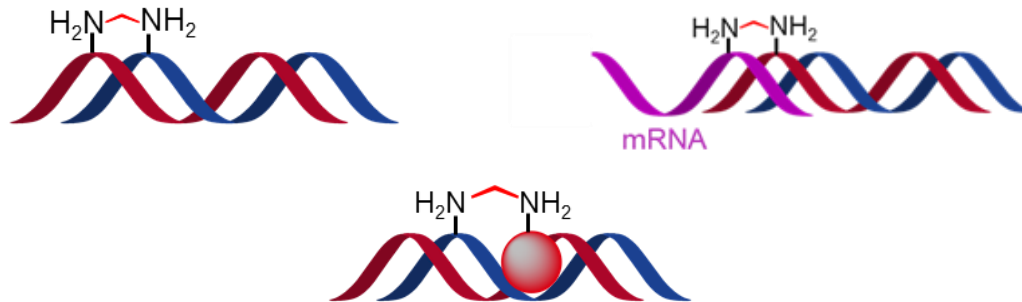
Methyl DNA Adducts



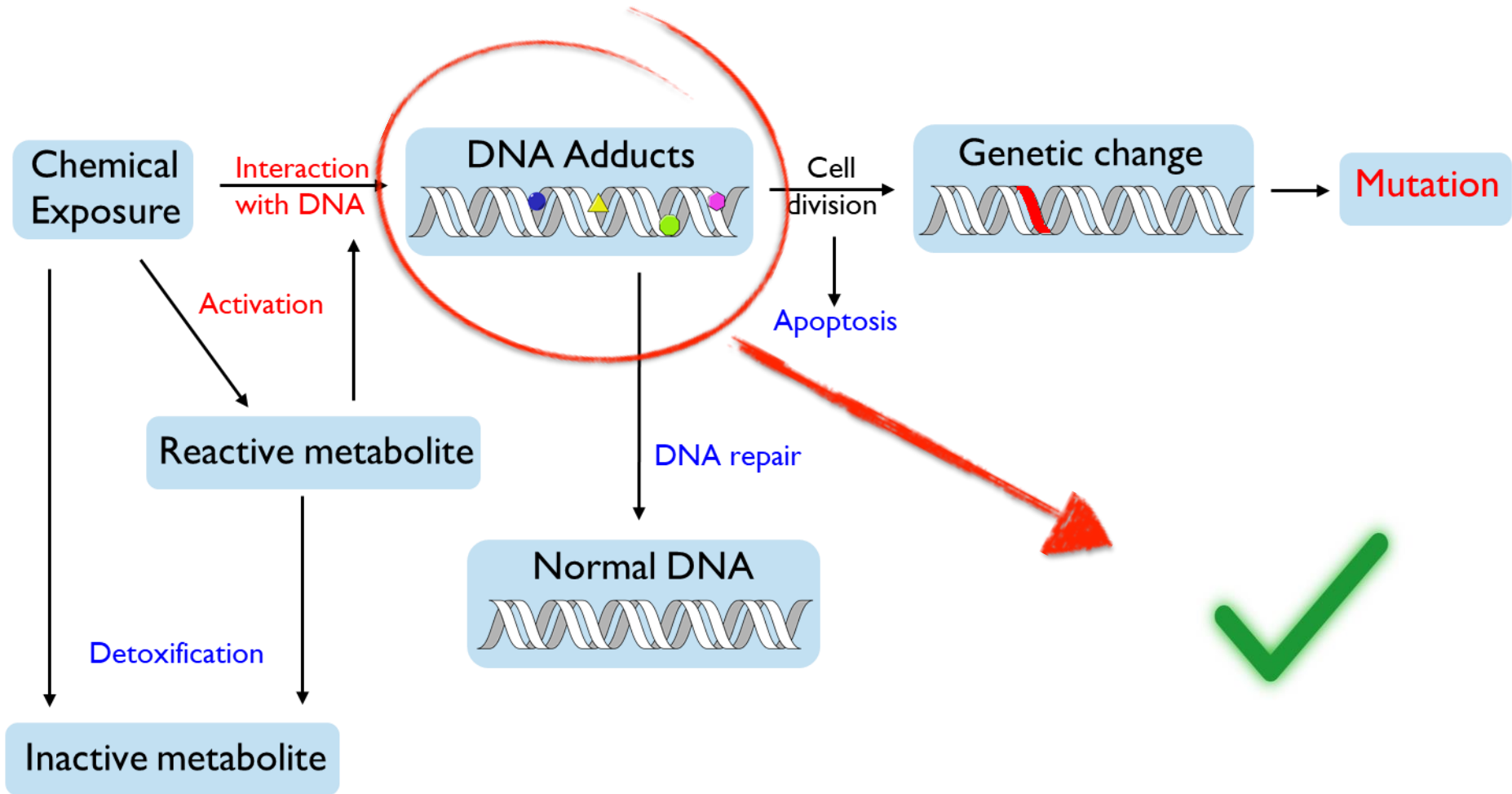
POB/PHB DNA Adducts



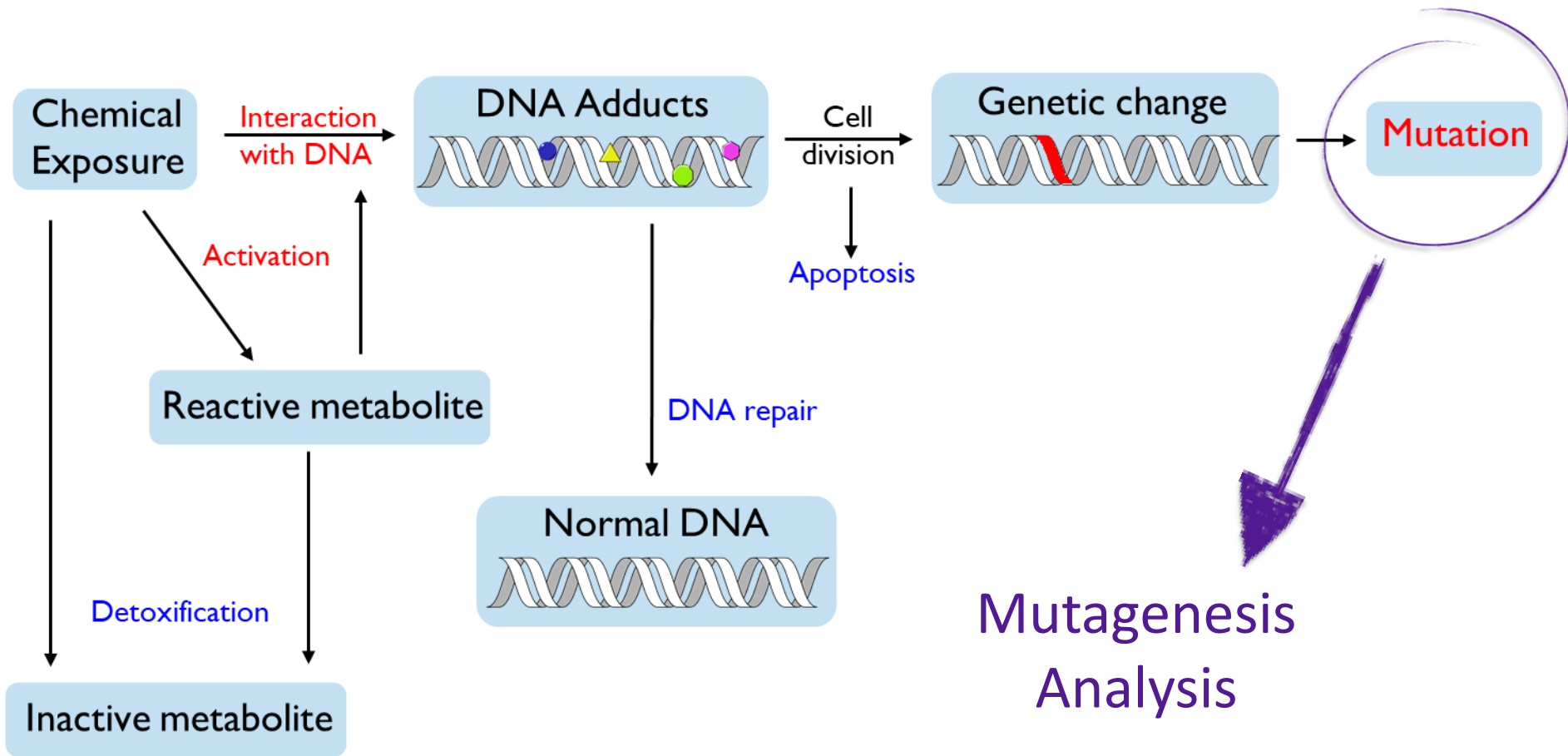
Cross-links (DNA-DNA, DNA-RNA, DNA-protein)



Chemical Carcinogenesis



Chemical Carcinogenesis



Sequencing sample preparation



DNA Extraction



Library Preparation



Duplex Sequencing

Sequencing sample preparation



DNA Extraction

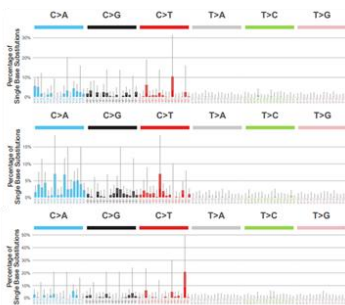


Library Preparation

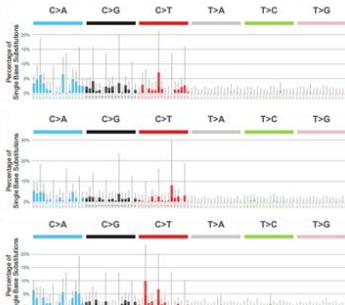


Duplex Sequencing

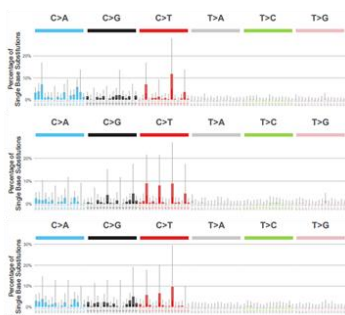
LPS 1 Week



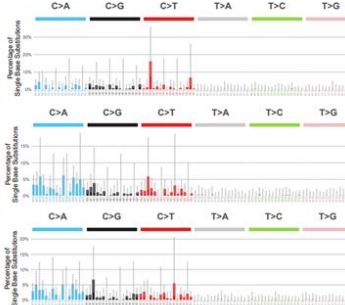
LPS 5 Weeks



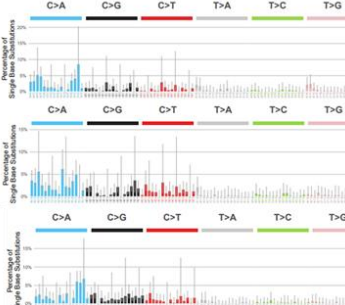
LPS 15 Weeks



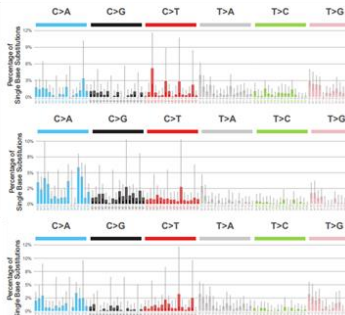
NNK+LPS 1 Week



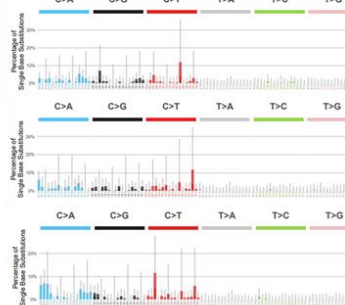
NNK+LPS 5 Weeks



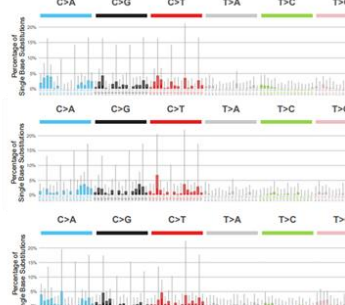
NNK+LPS 15 Weeks



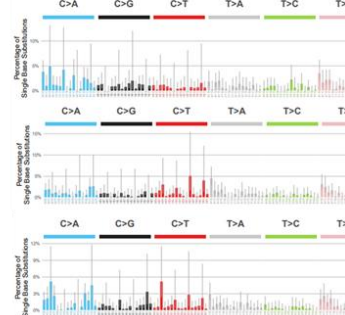
NNK 1 Week



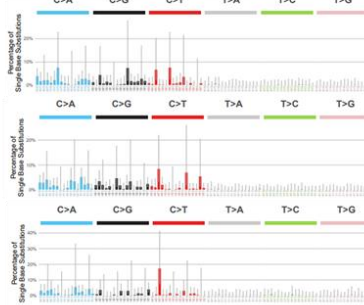
NNK 5 Weeks



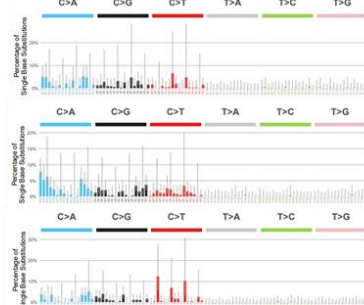
NNK 15 Weeks



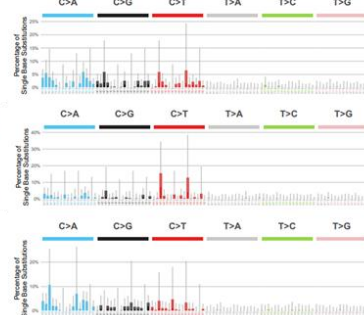
CTRL 1 Week



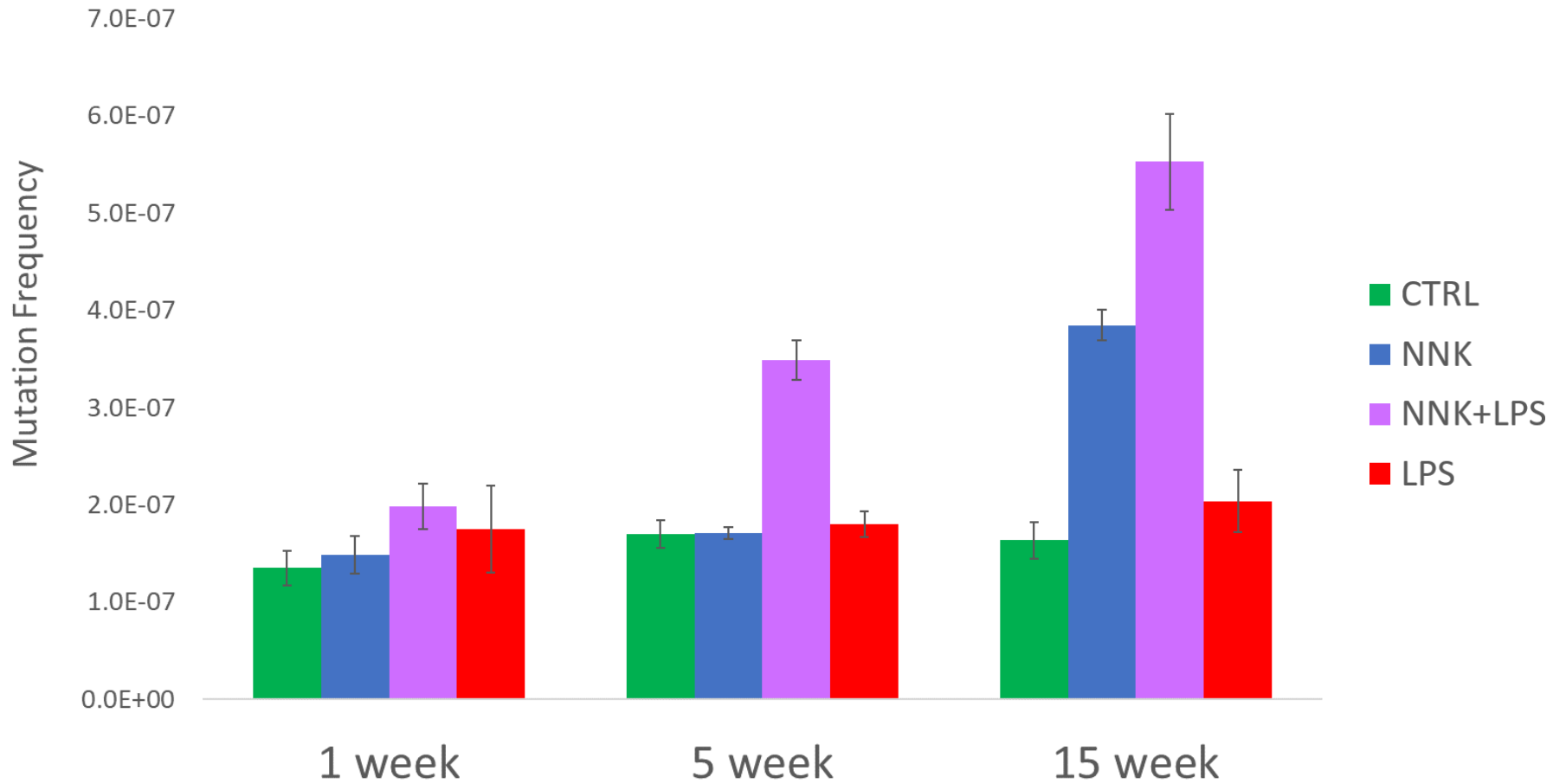
CTRL 5 Weeks



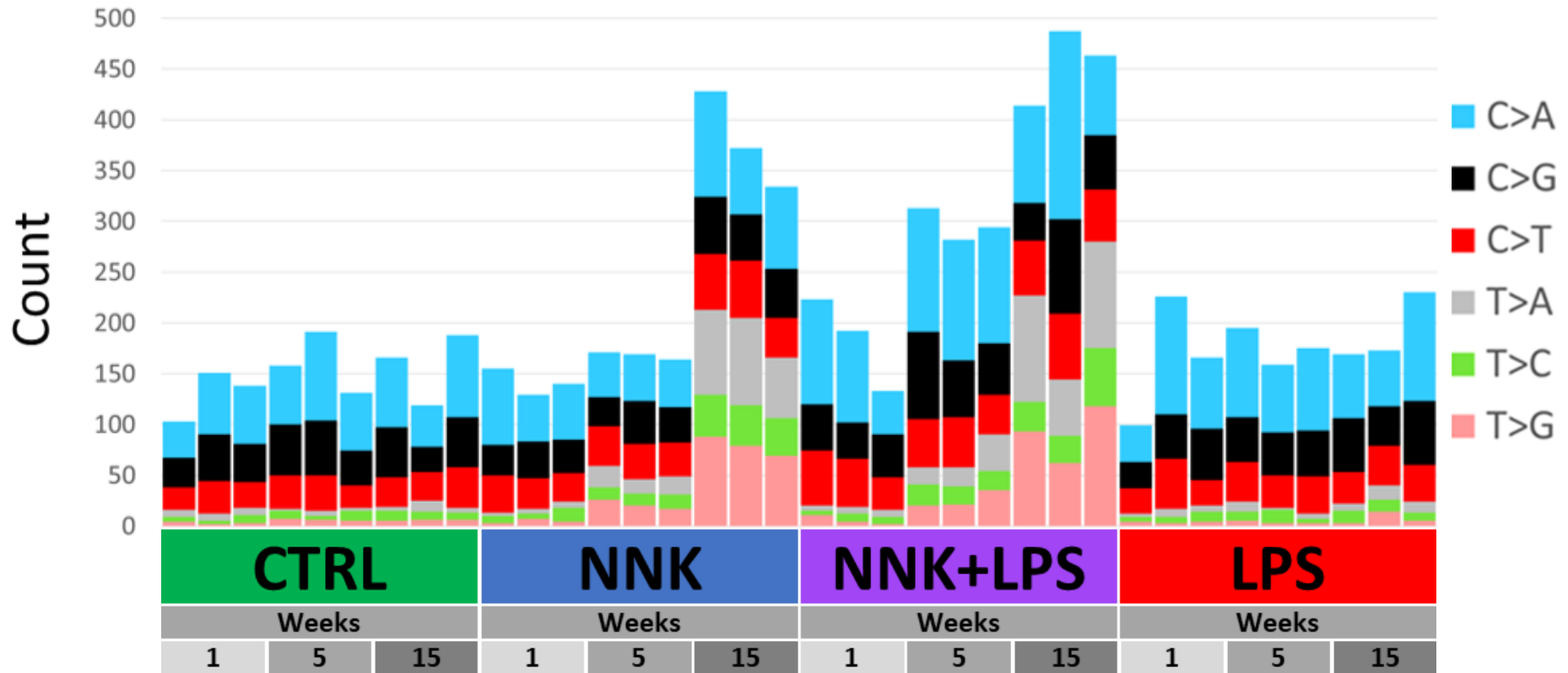
CTRL 15 Weeks



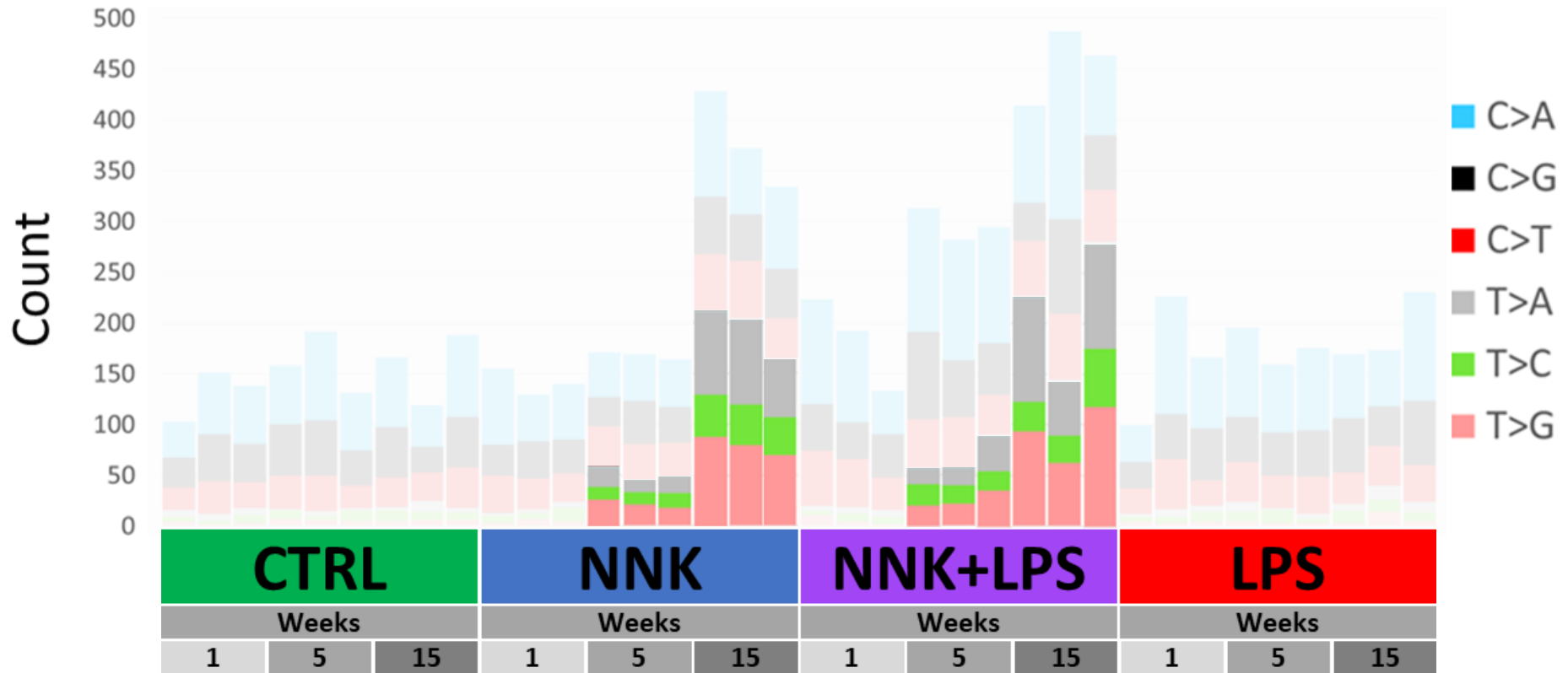
Mutation Frequency



SBS Mutation Counts



SBS Mutation Counts



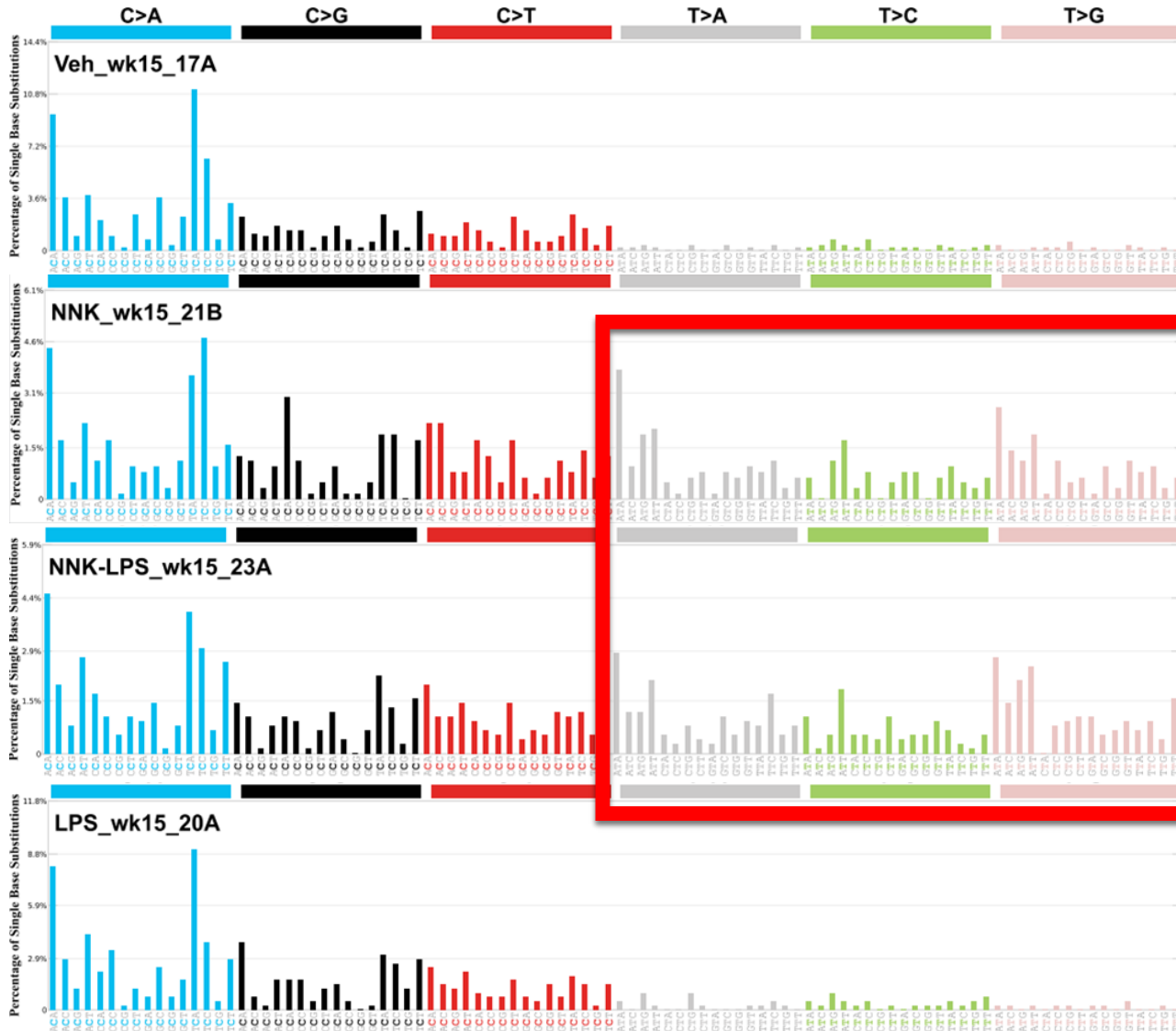
Trinucleotide Mutation Spectra

CTRL
15 Weeks

NNK
15 Weeks

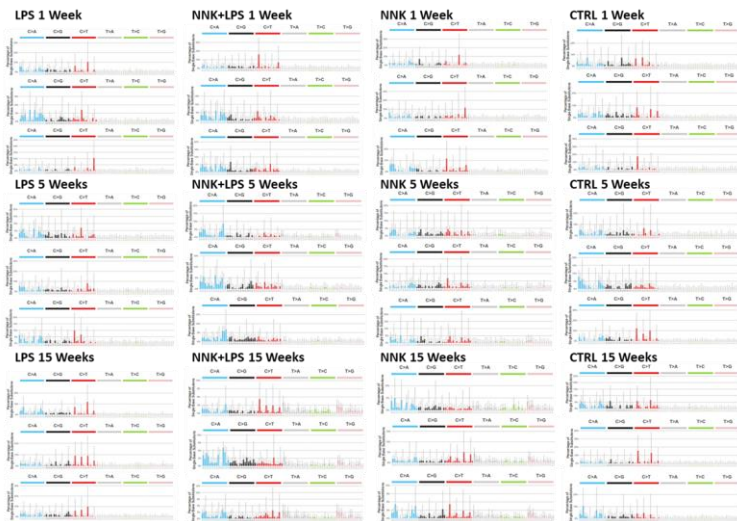
NNK + LPS
15 Weeks

LPS
15 Weeks

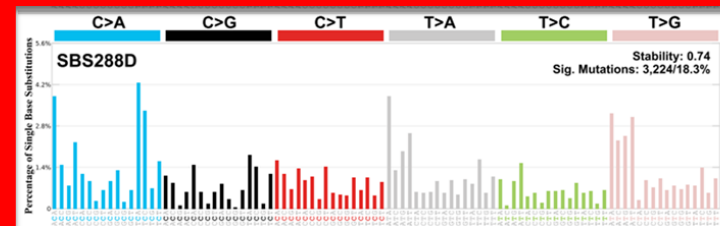
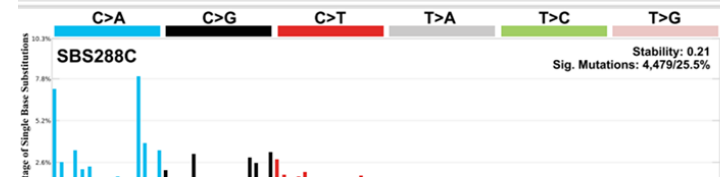
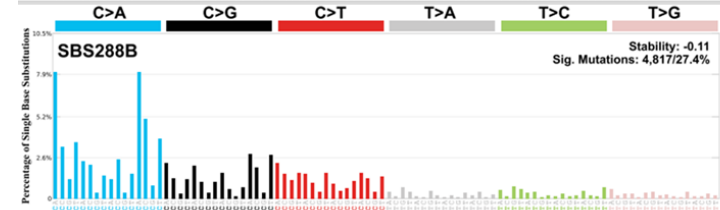
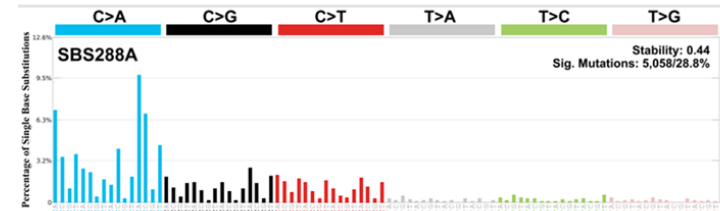


de novo Mutational Signatures

de novo mutational signatures

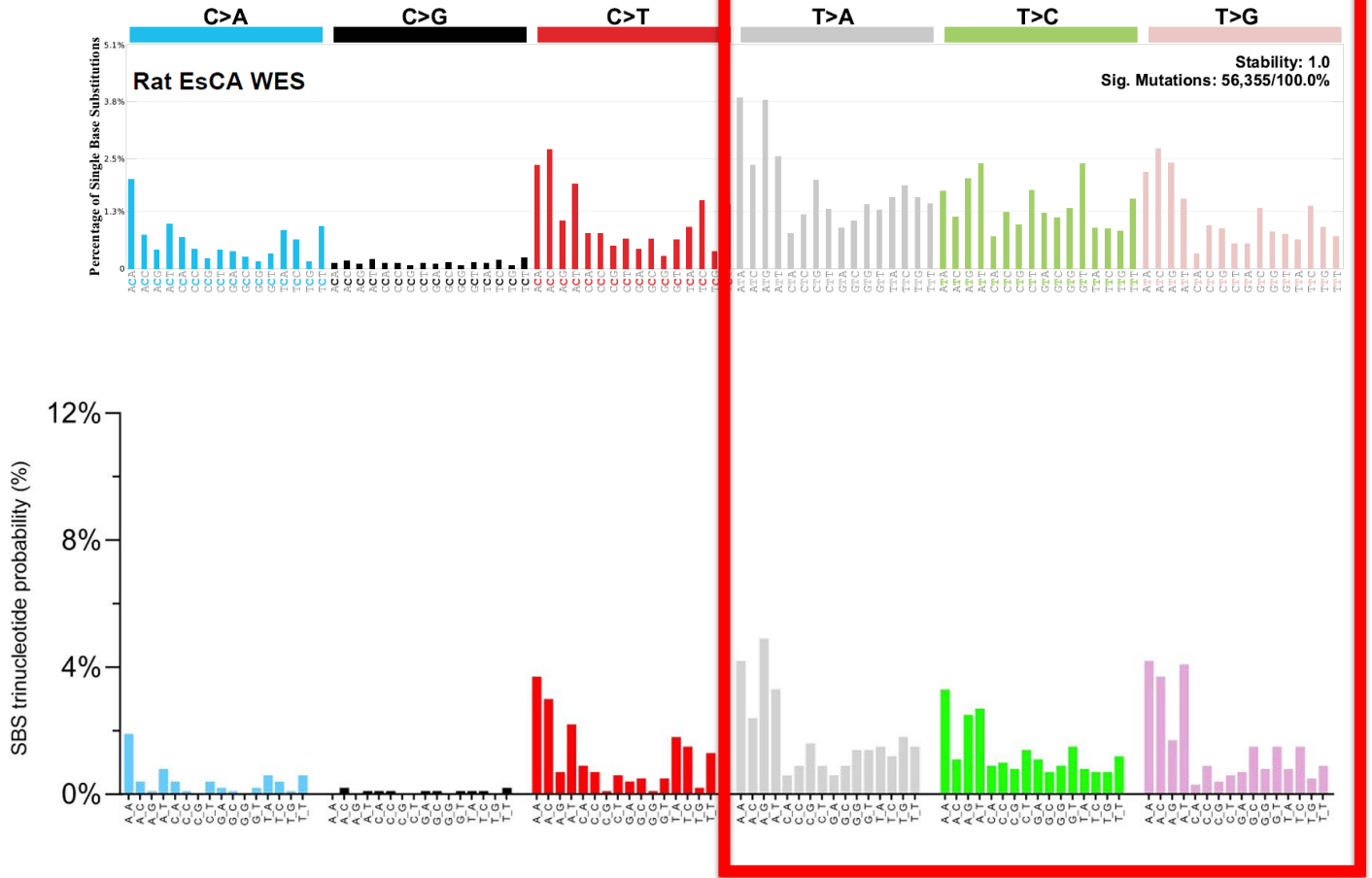


SIGPROFILER
Extractor

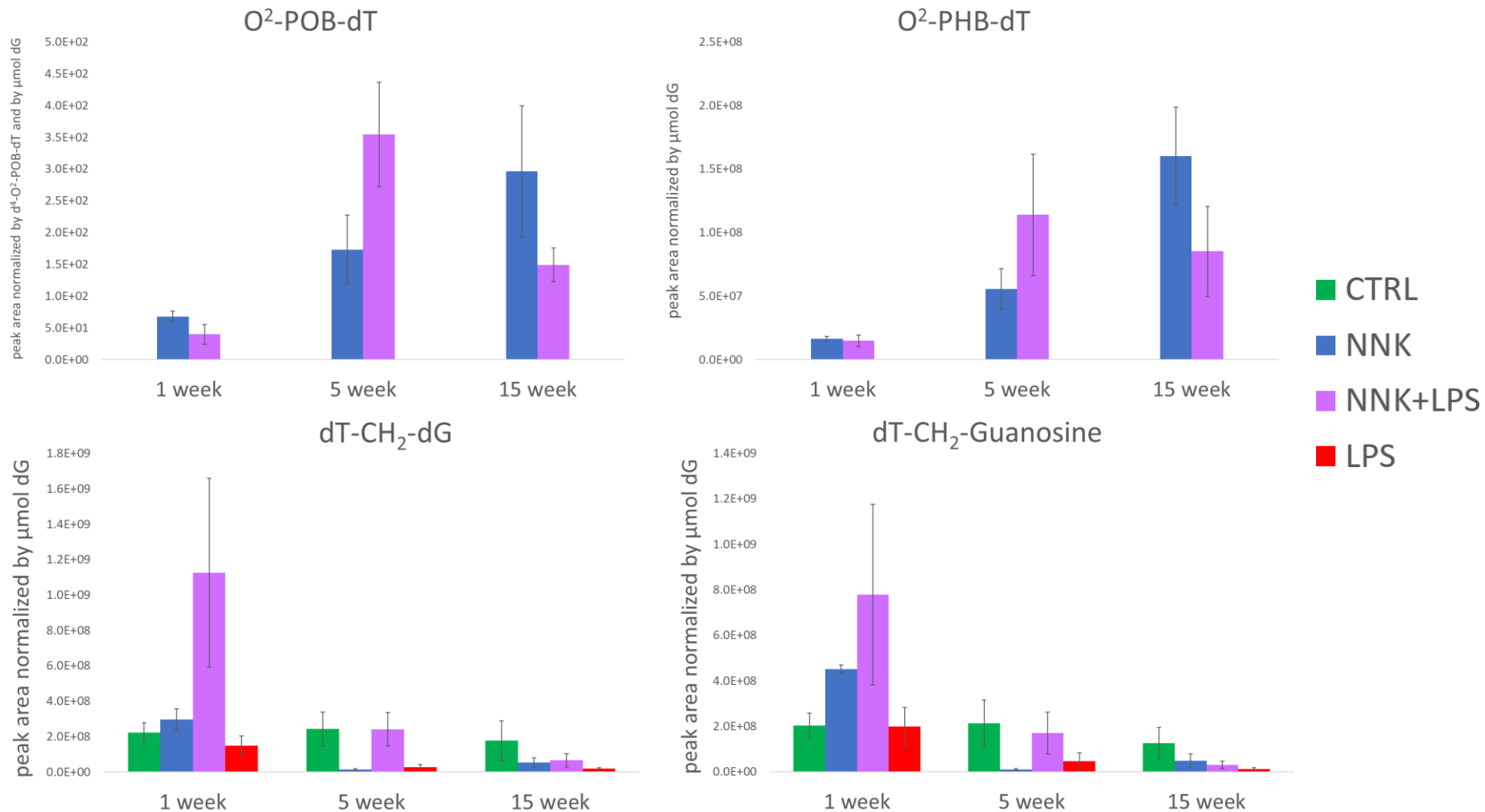


Recently reported NNK/NNN Mutational Signatures

- NNN-induced rat esophageal tumors
- BEAS-2B cells exposed to NNK acetate



Combining the adductome with the signatures



Key messages

- Exposure - Genotoxicity - Mutagenicity and Carcinogenicity are steps occurring at different times.
- Carcinogenesis is complex we need to keep in mind all mechanisms.
- We have methods to understand the chemical origins of DNA damage.
- We have methods to understand the consequences of genotoxicity.
- We have the opportunity to elucidate the steps of the carcinogenic process (including promotion) in ways that were not possible before.
- Exposure are often in combination and may be the result of complex mixtures.

Questions?

The case of Chronic Nephropathy and Urothelial Carcinoma of the Upper Urinary Tract



100 AAN/UUC....1800 at risk

THE JOURNAL OF ALTERNATIVE AND COMPLEMENTARY MEDICINE
Volume 4, Number 1, 1998, pp. 9-13
Mary Ann Liebert, Inc.

Misuse of Herbal Remedies: The Case of an Outbreak of Terminal Renal Failure in Belgium (Chinese Herbs Nephropathy)

JEAN-LOUIS VANHERWEGHEM, M.D., Ph.D.

ABSTRACT

At least 100 cases of extensive interstitial fibrosis of the kidneys were observed in Belgium in women who had followed a weight-loss regimen that included the use of Chinese herbs. The possible relation between the renal disease and these Chinese herbs was investigated. It was shown that the prescribed Chinese herb called *Stephania tetrandra* was, in fact, inadvertently replaced by another Chinese herb, namely *Aristolochia fangchi* in the powdered extracts delivered in Belgium and in France. The development of renal disease in about 100 patients exposed to the so-called *Stephania tetrandra* stresses the need for more stringent control of herbal medicine.

100 cases among women following a slimming diet based on consumption of a Chinese herbal remedy

The case of Chronic Nephropathy and Urothelial Carcinoma of the Upper Urinary Tract



Balkan endemic nephropathy.
Localized areas with high
incidence of the disease.

The central role of Aristolochia Herbs



100 AAN/UUC...1800 at risk



~100,000 at risk



~100,000,000 at risk

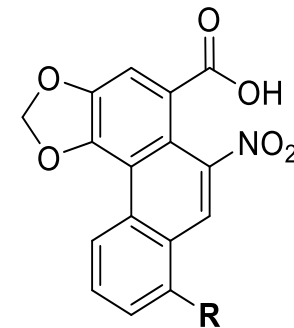
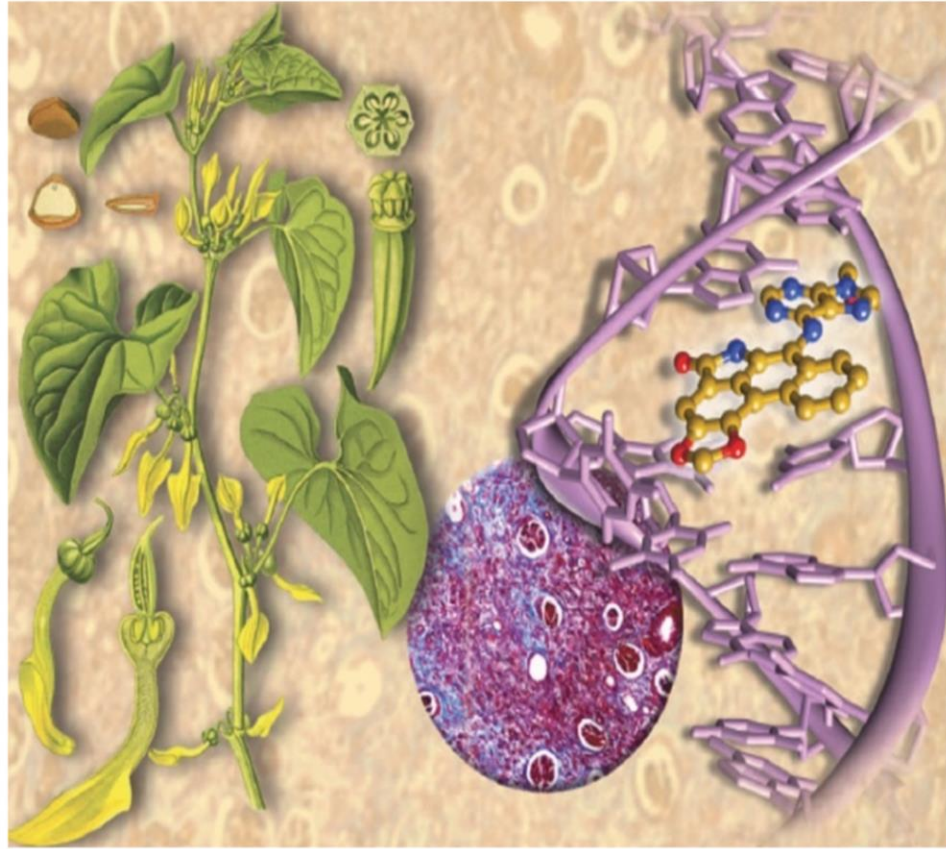


??



~8,000,000 at risk

Aristolochic Acid

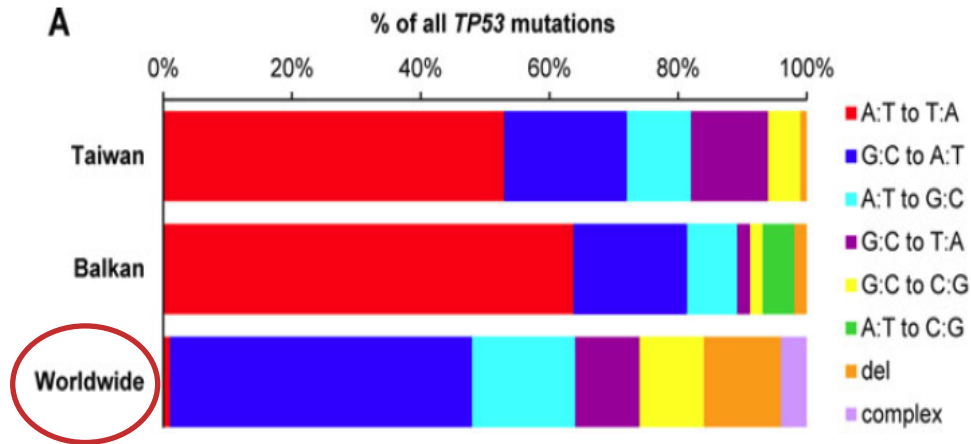


AA-I: R = OCH₃

AA-II: R = H

- Potent human carcinogen and nephrotoxin produced naturally by *Aristolochia* plants
- AA causes otherwise rare urothelial carcinomas of the upper urinary tract (UUC)
- AA is responsible for the otherwise uncommon A:T-to-T:A transversions in *P53* gene in patients with UUC

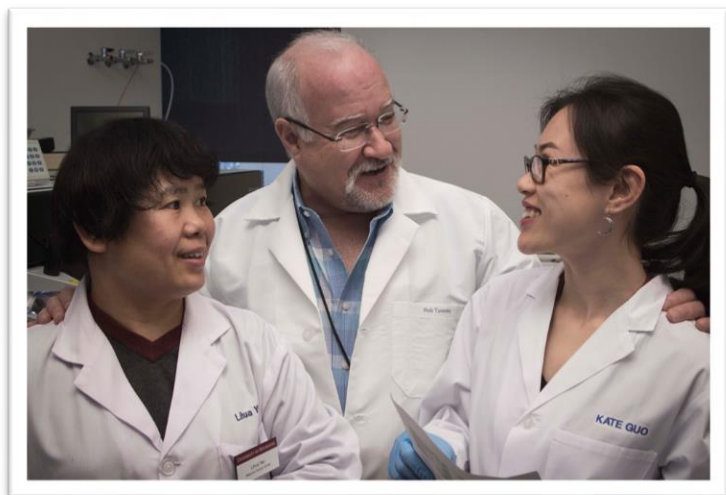
TP53 mutations



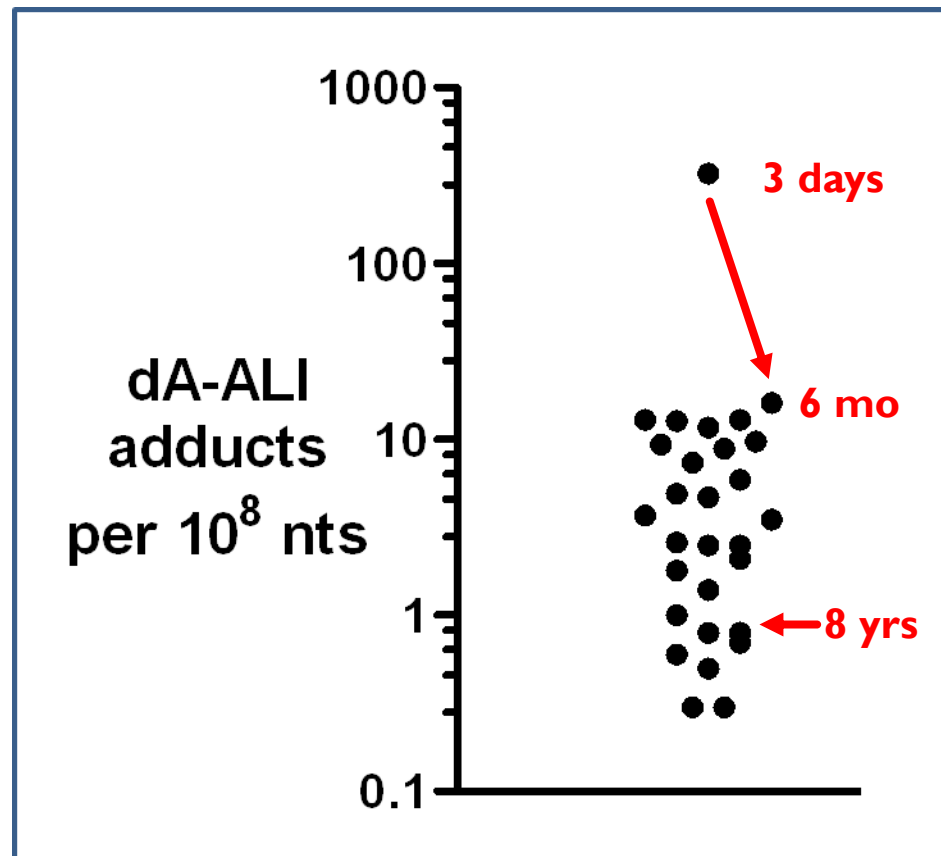
The AL-DNA adducts uniquely induce A to T transversion mutation in human *TP53* gene of patients with Aristolochic Acid Nephropathy (AAN, formerly Chinese herbs nephropathy, CHN) and Balkan Endemic Nephropathy (BEN)



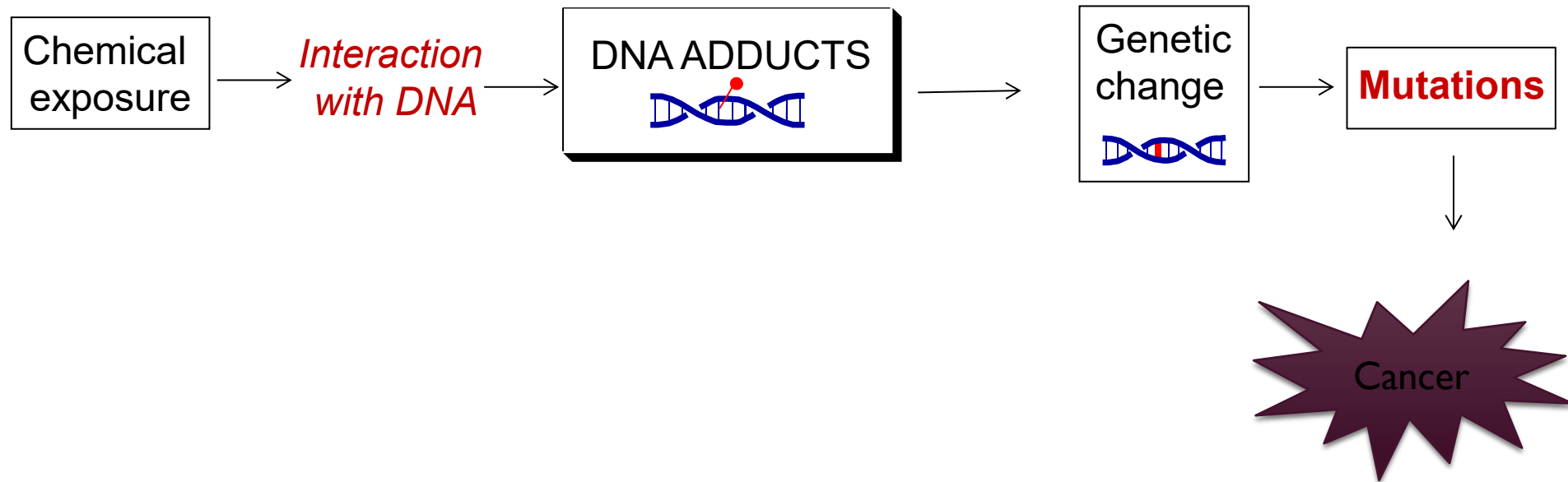
Noninvasive Biomonitoring DNA Adducts in Exfoliated Urinary Cells



32 samples from China
Clinical diagnosis of AAN
75% CKD stage 4/5
7 males/25 females
Time after withdrawal: 3d-8 yrs



CHEMICAL CARCINOGENESIS



TAKE HOME MESSAGES

- Just like defining exposure, defining the endpoint effect can be complex.
- Environmental and lifestyle-factors can induce DNA damage.
- Environmental and lifestyle-factors can induce cancer.
- In some cases an association is clear, but in most cases it is not, especially depending on the endpoint selected.
- Understanding these associations is crucial to identify opportunities for intervention and for prevention and for best use practices related to handling a chemical